JOINS OF LOCALISATIONS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: THE FINITE-ITERATION CRITERION FOR CLOSURE OPERATORS ON EXACT CATEGORIES, WITH APPLICATIONS TO TOPOSES, TORSION THEORIES AND FUZZY SETS

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Abstract. A pullback-stable, intersection-preserving closure operator c on an exact category determines the class W_c of c-bidense maps: for a Lawvere–Tierney topology the maps inverted by sheafification, for a hereditary torsion theory the maps inverted by the Gabriel quotient, for a nucleus the inequalities it identifies. In the lattice of classes closed under composition and two-out-of-three, the join $W_c \vee W_{c'}$ consists of the maps whose image and kernel pair are reached by boundedly many alternating applications of c and c', and in a Grothendieck topos or Grothendieck abelian category $W_c \vee W_{c'} = W_{\bar{c}}$, for $\bar{c}$ the least closure operator above both, iff $(cc')^k = \bar{c}$ for some k. It fails for persistence modules and for the path algebra of the quiver with two opposite arrows, and holds with $k = 1$ for stable torsion theories, so over commutative Noetherian rings. Alone, $k = 1$ does not give a two-step factorisation of the inverted maps, for topologies or torsion theories; in a topos it does when the second sheafification preserves the sheaves of the first, as for open, closed and double-negation topologies against every topology, and on a locale such topologies are characterised by their nuclei.

1. Introduction

Let $\mathcal{C}$ be a category. A *sameness* on $\mathcal{C}$ is a class of morphisms containing the identities and closed under composition and two-out-of-three; samenesses form a complete lattice $\mathrm{Same}(\mathcal{C})$ under inclusion, meets being intersections and the join of two classes the least sameness containing both [14]. Every localisation of $\mathcal{C}$, in particular every reflective subcategory, contributes the class W of maps it inverts. For a family of localisations closed under joins, the question studied in [16, 17, 23] is whether W sends the join of two localisations to the join of the two classes: whether every map inverted by the join localisation is obtained from maps inverted by the two

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localisations by finitely many compositions and cancellations. For subtoposes of a presheaf topos the answer was known to be yes on every finite category [17] and no on the poset $\mathbb{N}$ [16], and [23] gave the exact criterion for subtoposes of a Grothendieck topos: $W_j \vee W_{j'} = W_{j\vee j'}$ iff $(jj')^k = j \vee j'$ for some k, an identity between endomorphisms of the subobject classifier. The same identity, $(j_2 j_1)^N = j_1 \vee j_2$ for nuclei on a Heyting algebra, is the finite-iteration axiom of the lax logic of the join of two nuclei [19], and it is the exact condition for the join of the classes of isomorphisms up to the two nuclei in the sameness lattice of fuzzy sets to be the class of the join nucleus [21], whereas in the sameness lattice of the Heyting algebra itself the pointwise identity suffices.

This paper proves one theorem behind these results and applies it to a setting not covered by them, the Gabriel localisations of abelian categories. The common structure is a *closure operator* c on the subobjects of an exact category $\mathcal{C}$: a pullback-stable, intersection-preserving closure on each $\mathrm{Sub}(X)$. It determines the sameness W_c of *c-bidense* maps, the maps $f\colon X \to Y$ with $c(\mathrm{im} f) = Y$ and $\ker f \subseteq c(\Delta_X)$. For two closure operators c, c' let U_k be the class of maps with

$$(cc')^k(\mathrm{im} f) = Y \qquad \text{and} \qquad \ker f \subseteq (cc')^k(\Delta_X),$$

the maps that are bidense *uniformly* for the k-fold alternation of the two closures.

Theorem 1.1. *Let c, c' be closure operators on an exact category $\mathcal{C}$. Then*

$$W_c \vee W_{c'} = \bigcup_k U_k$$

in $\mathrm{Same}(\mathcal{C})$*; U_0 is the class of isomorphisms, and for $k \geq 1$ every map in U_k is a composite of $2k$ regular epimorphisms followed by $2k$ monomorphisms lying alternately in $W_{c'}$ and W_c. In particular the join is the closure of $W_c \cup W_{c'}$ under composition.*

Theorem 1.2. *Let $\mathcal{C}$ be a Grothendieck topos or a Grothendieck abelian category, and c, c' closure operators on $\mathcal{C}$. Then the least closure operator $\bar{c}$ above c and c' is the transfinite iterate of cc', and $W_c \vee W_{c'} = W_{\bar{c}}$ if and only if $(cc')^k = \bar{c}$ for some k.*

Theorem 1.2 is proved under axioms (Section 4) that isolate what is used: chains of subobjects have pullback-stable unions, and the *fold map* of two copies of an object glued along a subobject S witnesses whether S is reached in k steps; the two classes of categories named in the statement are the ones in which we verify the axioms, by different arguments.

In a Grothendieck topos the closure operators are the Lawvere–Tierney topologies and W_c is the class of maps inverted by sheafification; this is the main theorem of [23], whose proofs are reproduced here in the generality in which they were written, and whose consequences for presheaf toposes, ideal topologies, locales and subspaces we summarise (Section 5.1, 5.2). In

a Grothendieck abelian category $\mathcal{A}$ the closure operators are exactly the hereditary torsion theories T (Proposition 5.13): $c_T(A)$ is the preimage of the torsion part of X/A, W_{c_T} is the class of maps with kernel and cokernel in T, the maps inverted by the quotient functor $\mathcal{A} \to \mathcal{A}/T$, and $\bar{c}$ is the closure operator of the join torsion theory $T \vee T'$. Theorems 1.1 and 1.2 read:

Corollary 1.3. *Let T, T' be hereditary torsion theories on a Grothendieck abelian category $\mathcal{A}$ and let E_k be the class of objects having a filtration $0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_{2k} = M$ with $M_{2i+1}/M_{2i} \in T'$ and $M_{2i+2}/M_{2i+1} \in T$. Then $W_T \vee W_{T'}$ is the class of maps whose kernel and cokernel lie in $\bigcup_k E_k$, the closure of $W_T \cup W_{T'}$ under composition; and $W_T \vee W_{T'} = W_{T \vee T'}$ iff $T \vee T' = E_k$ for some k, i.e. iff the objects of the join torsion class have alternating T'/T-filtrations of bounded length.*

For persistence modules, i.e. representations $\cdots \to M_2 \to M_1 \to M_0$ of the ordinal ω in vector spaces, with T and T' the modules supported on the even and on the odd vertices, W_T is the class of maps that are isomorphisms at the odd vertices, $T \vee T'$ is everything, and the interval module $k[0, N]$ needs an alternating filtration of length $N + 2$; so $W_T \vee W_{T'}$ is a proper class, the class of maps whose kernel and cokernel are "bounded in alternation" (Example 5.14). This is the linear analogue of the failure for sheaves on $\mathbb{N}$ [16], and, as far as we know, the first statement of this kind about Gabriel localisations. Stability settles the question for the pairs that arise in commutative algebra. A hereditary torsion theory is *stable* if its torsion class is closed under injective hulls [22, Ch. VI, §7].

Theorem 1.4. *Let T, T' be hereditary torsion theories on a Grothendieck abelian category such that T' and $T \vee T'$ are stable. Then every object M of $T \vee T'$ has $M/t(M) \in T'$; hence $c_{T'} c_T = c_{T \vee T'}$, and*

$$W_T \vee W_{T'} = W_{T \vee T'} = W_{T'} \circ W_T,$$

every map $f \colon X \to Y$ inverted by the join localisation being the composite of a map inverted by Q_T followed by a map inverted by $Q_{T'}$, namely, once the T-torsion of the kernel is divided out, through the pullback of $Y \to Q_T(Y)$ along $Q_T(f)$ (Proposition 5.17). If T and T' are both stable, so is $T \vee T'$, and c_T and $c_{T'}$ commute.

The commutation concerns the closure operators on subobjects, not the radicals: the radicals of any two hereditary torsion theories commute, $t\,t'(M) = t(M) \cap t'(M) = t'\,t(M)$ by left exactness, whereas $c_T c_{T'}(A) = \pi^{-1} t(\pi'^{-1} t'(X/A))$ involves the torsion of successive quotients, and Example 5.14 shows that these closure operators need not commute. The ingredients of the proof, the decomposition of an injective object into its torsion part and a torsion-free complement for a stable theory, are classical [22, Ch. VI, §7], [6]; what is new is the consequence for the join of the inverted classes and its sharpness. Over a commutative Noetherian ring R every

hereditary torsion theory is stable and is the class T_Y of modules supported in a specialisation-closed subset Y of $\operatorname{Spec} R$ (Gabriel), and W_{T_Y} is the class of homomorphisms whose associated map of quasi-coherent sheaves is an isomorphism over the complement of Y. So $Y \mapsto W_{T_Y}$ is a lattice embedding of the specialisation-closed subsets of $\operatorname{Spec} R$ into $\operatorname{Same}(\operatorname{Mod} R)$, and a homomorphism of R-modules which is an isomorphism of sheaves over the complement of $Y \cup Y'$ is the composite of a homomorphism which is an isomorphism over the complement of Y followed by one which is an isomorphism over the complement of Y' (Corollary 5.20). Noetherianness alone does not suffice: for the path algebra of the quiver $1 \rightleftarrows 2$, a Noetherian ring, the torsion theories of the modules vanishing at one vertex are not stable, and the string modules need alternating filtrations of unbounded length (Example 5.21). On the topos side the counterpart of stability is a one-sided commutation: the identity $cc' = \bar{c}$ holds iff $c'c \leq cc'$ (Lemma 4.4), the open and the double-negation topologies satisfy it against every topology, the closed topologies satisfy the reverse one (Proposition 5.4), and in these three cases the maps inverted by the join sheafification factor in two steps through the pullback along one of the two sheafifications (Proposition 5.4). The identity alone does not give the factorisation, neither for topologies (Example 5.8) nor for torsion theories (Example 5.19), so that stability is needed in Theorem 1.4: in both examples the closure operators commute and the factorisation holds in one order only. In a topos the factorisation through the pullback along the c'-sheafification holds iff the c-sheafification preserves c'-sheaves, iff the c'-sheafification preserves c-bidense maps (Proposition 5.5); under $c'c \leq cc'$ it always preserves the c-dense monomorphisms (Proposition 5.6), and the c-bidense epimorphisms are where it can fail. On a frame these one-sided commutations with all nuclei characterise the nuclei with $n(x) \in \{y \to x\}$, which are the open ones when the frame is finite, and the closed nuclei (Proposition 5.11), and in a localic topos the identity $cc' = c \vee c'$ holds iff it holds for the nuclei, so that these are exactly the topologies satisfying it, respectively its reverse, against every topology (Corollary 5.12).

The frame L itself, viewed as a poset category, is an exact category in which every morphism is a monomorphism, with closure operators $c_n(x \leq y) = n(x) \wedge y$ for nuclei n; Theorem 1.1 applies and gives $W_n \vee W_{n'} = \{x \leq y : (nn')^k x \geq y \text{ for some } k\}$, so that the join is the class of the join nucleus iff the identity $(nn')^k x = (n \vee n')x$ holds *pointwise*; the witnesses of Theorem 1.2 do not exist in a poset, and indeed the uniform identity is not needed there (Section 5.3). Fuzzy sets are handled by a monomorphism version of Theorem 1.1 (Proposition 3.5), since the maps inverted by the reflection onto j-closed fuzzy sets are bijections; this recovers the description of their joins by bounded alternating chains [21] (Section 5.4).

The proofs are elementary and use only the calculus of closure operators on subobjects. The inclusion $U_k \subseteq W_c \vee W_{c'}$ is a peeling argument: a regular epimorphism whose kernel pair lies in the k-fold alternating closure of the

diagonal is divided by the equivalence relation "kernel pair intersected with the c'-closure of the diagonal", which is a c'-bidense map, and the induced map needs one closure fewer; dually a monomorphism is enlarged to the c'-closure of its image. The failure direction of Theorem 1.2 uses, for a $\bar{c}$-dense subobject S of Y that is not reached in k steps, the fold map of two copies of Y glued along S; the coproduct of these maps over all k is $\bar{c}$-bidense but lies in no U_k. The criterion is uniform in the subobject, and it is the iteration of the closure operators, not of the reflectors, that matters: on a monoid of order 8 the alternating sheafification tower never stabilises although the join is preserved (Remark 5.2).

The quotient-closed regime is different. For subvarieties R, R' of a variety of algebras, $W_R \vee W_{R'} = W_{R \cap R'}$ always holds, because the composite of the two reflectors is the reflector onto $R \cap R'$, while the meet $W_R \cap W_{R'}$ is the class of the join variety only under a fracture condition [18]; the natural closure operators there are not pullback-stable, and we explain in Section 6 why no common mechanism exists.

What is known. Samenesses and the join theorem for a composite reflector are from [14, 15]; the meet theorem, the reduction lemma and the $\mathbb{N}$ counterexample from [16]; the finite case from [17]; the criterion for subtoposes of Grothendieck toposes, with the presheaf, ideal-type, object-generated, localic and subspace cases, from [23]. Closure operators on categories are the subject of [3]; universal closure operators on a topos, their correspondence with Lawvere–Tierney topologies and the bidense description of the maps inverted by sheafification are in [13, V.1, V.3], [4, A4.3–A4.4]. Hereditary torsion theories, Gabriel's quotient categories and the characterisation of the maps they invert are in [5, 22]; the correspondence between hereditary torsion theories and closure operators on module categories is discussed in [3], and its extension to normal and homological categories in [2]; the join of hereditary torsion theories and its description by transfinite extensions go back to [9]. Joins of nuclei as transfinite iterates are in [10]; the weak topologies of [12] compute the join of two Lawvere–Tierney topologies as such an iterate. The lax logic of the join of two nuclei and its finite-iteration axiom are in [19]; the sameness lattice of fuzzy sets and the uniform and pointwise criteria in [21]; the quotient-closed regime in [18].

What is new. The main new result is Corollary 1.3: for hereditary torsion theories T, T' on a Grothendieck abelian category, the join of the classes of maps inverted by the two Gabriel quotients is the class of maps whose kernel and cokernel have finite alternating T'/T-filtrations, it is generated by composition alone, and it is the class of the join torsion theory exactly when the lengths are bounded; Example 5.14 shows that the bound can fail even when the join torsion theory is the whole category. The description of $T \vee T'$ by transfinite alternating filtrations and the dictionary between closure operators and hereditary torsion theories (Proposition 5.13) are classical; the statement about the join of the inverted classes, and its failure,

are new as far as we know. Theorem 1.4 rests on the classical decomposition of injectives for stable torsion theories [22, Ch. VI, §7], [6, 7], and the commutation of the closure operators is a direct consequence of it; the new content is that stable pairs satisfy the criterion with $k = 1$, the resulting two-step factorisation of the maps inverted by the join localisation through the pullback along the first localisation (Proposition 5.17), the embedding of the specialisation-closed subsets of $\operatorname{Spec} R$ into $\operatorname{Same}(\operatorname{Mod} R)$ for R commutative Noetherian (Corollary 5.20), and the Noetherian counterexample (Example 5.21). On the topos side, Proposition 5.5 characterises the two-step factorisation through the pullback along the c'-sheafification by the preservation of c'-sheaves under the c-sheafification, equivalently of c-bidense maps under the c'-sheafification, Proposition 5.6 shows that the latter preserves c-dense monomorphisms whenever $c'c \leq cc'$, and Example 5.8 shows that commuting topologies need not have the factorisation, as Example 5.19 shows for torsion theories; the particular cases are Proposition 5.4, with the characterisation of Proposition 5.11 and its localic form, Corollary 5.12, which decides the identity $cc' = c \vee c'$ on a localic topos at the level of the nuclei. The general Theorems 1.1 and 1.2 are the tool: Theorem 1.1 is the peeling argument of [23, Thm. A] carried out in an exact category instead of a topos, and Theorem 1.2 replaces the subobject classifier used in [23, Thm. B] by the witness and sum axioms, which is what makes the abelian case accessible; neither adds anything in the topos case, and their contribution is the generality; the topos-theoretic, localic, frame and fuzzy-set statements of [23, 21] are recovered as instances, with the pointwise/uniform difference explained by the presence or absence of witnesses (Section 5), and the quotient-closed regime is contrasted in Section 6.

2. Closure operators on exact categories

Throughout, $\mathcal{C}$ is an *exact* category in the sense of Barr [1]: it has finite limits, every morphism factors as a regular epimorphism followed by a monomorphism, regular epimorphisms are stable under pullback, and every equivalence relation is the kernel pair of its coequaliser. We write $\operatorname{im} f \subseteq Y$ for the image of $f\colon X \to Y$, $\ker f \subseteq X \times X$ for its kernel pair, $\Delta_X \subseteq X \times X$ for the diagonal, $g^{-1}(B)$ for the pullback of a subobject and $g(A)$ for the image of a subobject. We use freely the calculus of subobjects in a regular category: $\operatorname{im}(gh) = g(\operatorname{im} h)$; $\ker(gh) = (h \times h)^{-1}(\ker g)$; for a regular epimorphism e, $e(e^{-1}(B)) = B$; products of regular epimorphisms are regular epimorphisms; and for $g\colon Y \to Z$ and $A \subseteq Y$,

$$g^{-1}(g(A)) = \pi_1\big(\ker g \cap (Y \times A)\big),$$

as both are the image of $Y \times_Z A \to Y$.

Definition 2.1. A *closure operator* on $\mathcal{C}$ assigns to each object X a map $A \mapsto c(A)$ on $\operatorname{Sub}(X)$ that is inflationary, monotone, idempotent, preserves binary intersections, and is stable under pullback: $g^{-1}(c(B)) = c(g^{-1}(B))$

for every $g\colon X \to Y$ and $B \subseteq Y$. Closure operators are ordered pointwise. A subobject $A \subseteq X$ is *c-dense* if $c(A) = X$.

These are idempotent closure operators in the sense of [3], with the additional requirements that they preserve binary intersections and commute with pullbacks along all morphisms (the hereditary property of [3] concerns pullbacks along monomorphisms only). On a topos they are exactly the universal closure operators, which correspond to the Lawvere–Tierney topologies [13, V.1]; on a Grothendieck abelian category they are exactly the hereditary torsion theories (Proposition 5.13). A *word* is a composite $w = c_1 c_2 \cdots c_n$ of closure operators, c_n applied first; words are inflationary, monotone, pullback-stable and preserve binary intersections, but need not be idempotent.

Lemma 2.2. *Let w be a word, $g\colon X \to Y$, $A, K \subseteq X$, $B \subseteq Y$.*

(i) $g(w(A)) \subseteq w(g(A))$.
(ii) $w(A) \cap K \subseteq w(A \cap K)$.
(iii) $w(A \times B) = w(A) \times w(B)$ *in* $\mathrm{Sub}(X \times Y)$.
(iv) *For a closure operator c, $c(\Delta_X)$ is an equivalence relation on X.*

Proof. It suffices to treat a single closure operator c in (i)–(iii) and to iterate. (i) $A \subseteq g^{-1}(g(A))$, so $c(A) \subseteq c(g^{-1}(g(A))) = g^{-1}(c(g(A)))$. (ii) Let $m\colon K \hookrightarrow X$; then $A \cap K = m(m^{-1}(A))$ and $c(A) \cap K = m(m^{-1}(c(A))) = m(c_K(m^{-1}(A))) \subseteq c(m(m^{-1}(A)))$ by (i), where c_K is the closure on $\mathrm{Sub}(K)$, the pullback of c. (iii) $A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B)$, and c preserves intersections and pullbacks. (iv) Reflexivity and symmetry are clear (the twist map preserves Δ_X and commutes with c). The composite of the relation $c(\Delta_X)$ with itself is $\pi_{13}\big(\pi_{12}^{-1}c(\Delta_X) \cap \pi_{23}^{-1}c(\Delta_X)\big) = \pi_{13}\big(c(\pi_{12}^{-1}\Delta_X \cap \pi_{23}^{-1}\Delta_X)\big) \subseteq c(\pi_{13}(\delta)) = c(\Delta_X)$ by (i), where $\delta \subseteq X^3$ is the image of the triple diagonal. □

Definition 2.3. A map $f\colon X \to Y$ is *c-bidense* if $\mathrm{im} f$ is c-dense in Y and $\ker f \subseteq c(\Delta_X)$. Write W_c for the class of c-bidense maps.

A monomorphism is c-bidense iff it is c-dense, and a regular epimorphism iff its kernel pair lies in $c(\Delta_X)$. On a topos with the closure operator of a Lawvere–Tierney topology j, W_c is the class of maps inverted by the sheafification a_j [4, A4.3]; in general W_c is defined by the closure operator alone, and Lemma 3.2 below (with $c' = c$) shows that it is a sameness.

3. Uniform bidensity and the join

Fix closure operators c, c' on $\mathcal{C}$ and write $c^{(k)} := (cc')^k$, a word of length $2k$ ending with c', $c^{(0)}$ being the identity; since closures are inflationary, $c^{(k)} \leq c^{(k+1)}$ and $c, c' \leq c^{(1)}$. Write $V := W_c \vee W_{c'}$.

Definition 3.1. For $k \geq 0$ let U_k be the class of maps $f\colon X \to Y$ with $c^{(k)}(\mathrm{im} f) = Y$ and $\ker f \subseteq c^{(k)}(\Delta_X)$. Put $U = \bigcup_k U_k$.

U_0 is the class of isomorphisms: $\operatorname{im} f = Y$ means that the monic part of f is an isomorphism, and $\ker f = \Delta_X$ that f is a monomorphism, so f is a monic regular epimorphism.

Lemma 3.2. *$U_k \subseteq U_{k+1}$, $W_c \cup W_{c'} \subseteq U_1$, and U is a sameness. Hence $V \subseteq U$. With $c' = c$ the same proof shows that W_c is a sameness.*

Proof. The first two assertions follow from $c^{(k)} \leq c^{(k+1)}$ and $c, c' \leq c^{(1)}$. Identities lie in U_0. Let $h\colon X \to Y$ and $g\colon Y \to Z$; we use Lemma 2.2 for the words $c^{(k)}$.

Composition. Let $h \in U_k$, $g \in U_l$. Then $\operatorname{im}(gh) = g(\operatorname{im} h)$ and $c^{(k+l)}(\operatorname{im} gh) \supseteq c^{(l)}\big(g(c^{(k)}(\operatorname{im} h))\big) = c^{(l)}(g(Y)) = c^{(l)}(\operatorname{im} g) = Z$ by (i). And $\ker(gh) = (h \times h)^{-1}(\ker g) \subseteq (h \times h)^{-1}(c^{(l)}(\Delta_Y)) = c^{(l)}((h \times h)^{-1}\Delta_Y) = c^{(l)}(\ker h) \subseteq c^{(l+k)}(\Delta_X)$.

Cancellation of g. Let $gh \in U_k$ and $g \in U_k$. Then $\ker h \subseteq \ker(gh) \subseteq c^{(k)}(\Delta_X)$. Let $B := g^{-1}(\operatorname{im} gh) \subseteq Y$; then $c^{(k)}(B) = g^{-1}(c^{(k)}(\operatorname{im} gh)) = g^{-1}(Z) = Y$. Moreover $B = \pi_1\big(\ker g \cap (Y \times \operatorname{im} h)\big)$, and

$$\begin{aligned}\pi_1\big(\ker g \cap (Y \times \operatorname{im} h)\big) &\subseteq \pi_1\big(c^{(k)}(\Delta_Y) \cap (Y \times \operatorname{im} h)\big)\\ &\subseteq \pi_1\big(c^{(k)}(\Delta_Y \cap (Y \times \operatorname{im} h))\big)\\ &\subseteq c^{(k)}\big(\pi_1(\Delta_Y \cap (Y \times \operatorname{im} h))\big) = c^{(k)}(\operatorname{im} h)\end{aligned}$$

by (ii) and (i). Hence $Y = c^{(k)}(B) \subseteq c^{(2k)}(\operatorname{im} h)$.

Cancellation of h. Let $gh \in U_k$ and $h \in U_k$. Then $\operatorname{im} g \supseteq \operatorname{im}(gh)$, so $c^{(k)}(\operatorname{im} g) = Z$. Since $h \times h\colon X \times X \to \operatorname{im} h \times \operatorname{im} h$ is a regular epimorphism, $\ker g \cap (\operatorname{im} h \times \operatorname{im} h) = (h \times h)(\ker gh) \subseteq (h \times h)(c^{(k)}(\Delta_X)) \subseteq c^{(k)}(\Delta_Y)$ by (i); and $c^{(k)}(\operatorname{im} h \times \operatorname{im} h) = Y \times Y$ by (iii). So $\ker g = \ker g \cap c^{(k)}(\operatorname{im} h \times \operatorname{im} h) \subseteq c^{(k)}\big(\ker g \cap (\operatorname{im} h \times \operatorname{im} h)\big) \subseteq c^{(2k)}(\Delta_Y)$ by (ii).

For $c' = c$ one has $c^{(k)} = c$ for all $k \geq 1$, so $U_k = W_c$ and the argument shows that W_c is closed under composition and cancellation. □

Lemma 3.3 (peeling)**.** *Let $w = c_1 c_2 \cdots c_n$ be a word in closure operators.*

(a) *If $f\colon X \to Y$ is a regular epimorphism with $\ker f \subseteq w(\Delta_X)$, then $f = f_1 \circ \cdots \circ f_n \circ u$ with u an isomorphism and f_i a regular epimorphism in W_{c_i}.*

(b) *If $f\colon X \to Y$ is a monomorphism with $w(X) = Y$, then $f = f_1 \circ \cdots \circ f_n \circ u$ with u an isomorphism and f_i a monomorphism in W_{c_i}.*

For $n \geq 1$ the isomorphism u can be absorbed into f_n.

Proof. Induction on n; for $n = 0$ the hypothesis says that f is an isomorphism. Write $w = w'c_n$. (a) Let $R := \ker f \cap c_n(\Delta_X)$, an equivalence relation on X by Lemma 2.2(iv), and let $q\colon X \twoheadrightarrow X/R$ be its coequaliser, which has $\ker q = R$ by exactness; so $\ker q \subseteq c_n(\Delta_X)$ and $q \in W_{c_n}$. Since $R \subseteq \ker f$, f coequalises the two projections of R and $f = \bar{f} \circ q$ with $\bar{f}$ a regular epimorphism, and $\ker \bar{f} = (q \times q)(\ker f)$ because $q \times q$ is a regular

epimorphism and $\ker f = (q \times q)^{-1}(\ker \bar{f})$. Now

$$\begin{aligned}(q \times q)(\ker f) &= (q \times q)\big(w'(c_n\Delta_X) \cap \ker f\big) \subseteq (q \times q)\big(w'(c_n\Delta_X \cap \ker f)\big) \\ &= (q \times q)(w'(R)) \subseteq w'\big((q \times q)(R)\big) = w'(\Delta_{X/R})\end{aligned}$$

by Lemma 2.2(ii) and (i), as $(q \times q)(\ker q) = \Delta_{X/R}$. Induction applied to $\bar{f}$ and w' gives the factorisation. (b) Regard X as a subobject of Y and let $Z := c_n(X) \subseteq Y$. The inclusion $X \hookrightarrow Z$ is c_n-dense (the closure of X in Z is $c_n(X) \cap Z = Z$ by pullback stability), hence in W_{c_n}, and $w'(Z) = w(X) = Y$. Induction applied to $Z \hookrightarrow Y$ and w' gives the factorisation. □

Proof of Theorem 1.1. Let C be the closure of $W_c \cup W_{c'}$ under composition. Then $C \subseteq V \subseteq U$ by Lemma 3.2. Let $f \in U_k$, $k \geq 1$, and factor $f = m \circ e$ with $e\colon X \twoheadrightarrow \mathrm{im}f$ a regular epimorphism and $m\colon \mathrm{im}f \hookrightarrow Y$ a monomorphism. Then $\ker e = \ker f \subseteq c^{(k)}(\Delta_X)$ and $c^{(k)}(\mathrm{im}f) = Y$, and $c^{(k)}$ is a word of length $2k$ ending with c'. By Lemma 3.3, e is a composite of $2k$ regular epimorphisms and m a composite of $2k$ monomorphisms, alternately in $W_{c'}$ and W_c, beginning with $W_{c'}$. So $U \subseteq C$, and $C = V = U$. □

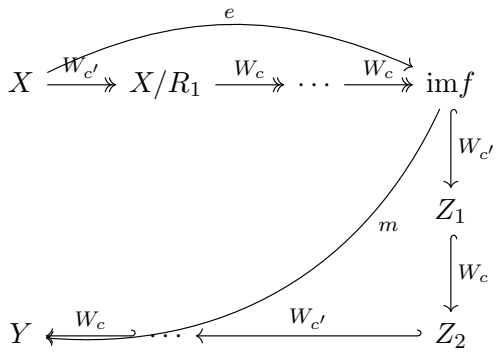


FIGURE 1. Theorem 1.1: a map in U_k is $2k$ quotients by "kernel pair intersected with the closure of the diagonal" followed by $2k$ alternating closures of its image, each step bidense for one of the two closure operators.

Remark 3.4. In general the join of two samenesses requires the two-out-of-three saturation of the compositional closure [14]: on the poset $0 < 1 < 2$ viewed as a category, the classes $W_1 = \{0 \to 1\}$ and $W_2 = \{0 \to 2\}$ (with the identities) are samenesses, their compositional closure is $\{0 \to 1, 0 \to 2\}$, and two-out-of-three forces $1 \to 2$ into the join, which is therefore the class of all morphisms. For two bidense classes no cancellation is needed. The number of steps is controlled in both directions: a composite of n maps of $W_c \cup W_{c'}$ lies in U_n, and a map of U_k is a composite of $4k$ such maps.

Proposition 3.5 (monomorphisms)**.** *Let $\mathcal{C}$ be a category with finite limits and c, c' closure operators on $\mathcal{C}$ (pullback-stable, intersection-preserving closures on the subobject lattices); exactness is not assumed. For monomorphisms $m\colon X \hookrightarrow Y$ let M_k be the class of those with $c^{(k)}(X) = Y$ and $M = \bigcup_k M_k$. Then M contains the c-dense and the c'-dense monomorphisms, is closed under composition, and under cancellation among monomorphisms: if $gh \in M_k$ then $h \in M_k$ and $g \in M_k$; and every $m \in M_k$ is a composite of $2k$ monomorphisms alternately c'-dense and c-dense. Consequently, if B is a sameness consisting of monomorphisms such that for $m\colon X \hookrightarrow Y$ in B the inclusions $X \hookrightarrow c(X)$ and $X \hookrightarrow c'(X)$ are in B, then the join in* $\mathrm{Same}(\mathcal{C})$ *of the c-dense and the c'-dense members of B is $M \cap B$.*

Proof. For $h\colon X \hookrightarrow Y$ in M_k and $g\colon Y \hookrightarrow Z$ in M_l, pullback stability along $Y \hookrightarrow Z$ gives $c_Z^{(k)}(X) \cap Y = c_Y^{(k)}(X) = Y$, so $c_Z^{(k)}(X) \supseteq Y$ and $c_Z^{(k+l)}(X) \supseteq c_Z^{(l)}(Y) = Z$. If $gh \in M_k$, then $c_Y^{(k)}(X) = c_Z^{(k)}(X) \cap Y = Z \cap Y = Y$, so $h \in M_k$, and $c_Z^{(k)}(Y) \supseteq c_Z^{(k)}(X) = Z$, so $g \in M_k$; neither cancellation needs the other factor to be in M. The factorisation is Lemma 3.3(b), whose proof uses only pullback stability. For the last assertion, $M \cap B$ is a sameness containing the two classes, and each $m \in M_k \cap B$ is a composite of its peeling factors, which lie in B by hypothesis and by two-out-of-three in B. □

4. The finite-iteration criterion

In this section $\mathcal{C}$ satisfies

(H1) every $\mathrm{Sub}(X)$ is a set and has unions of chains, these are stable under pullback, and $A \cap \bigcup_\alpha B_\alpha = \bigcup_\alpha (A \cap B_\alpha)$ for chains (B_α).

Let c, c' be closure operators and define the transfinite iterates $c^{(0)}(A) = A$, $c^{(\alpha+1)}(A) = c^{(1)}(c^{(\alpha)}(A))$ and $c^{(\lambda)}(A) = \bigcup_{\alpha<\lambda} c^{(\alpha)}(A)$ for limit λ; they increase with α, are constant from some ordinal γ_A on since $\mathrm{Sub}(X)$ is a set, commute with pullback by (H1), and $c^{(\alpha)}(c^{(\beta)}(A)) = c^{(\beta+\alpha)}(A)$.

Lemma 4.1. *Under (H1), $\bar{c}(A) := \bigcup_\alpha c^{(\alpha)}(A)$ is a closure operator, the least one above c and c'.*

Proof. $\bar{c}$ is inflationary and monotone; $\bar{c}(A) = c^{(\gamma)}(A)$ for $\gamma = \gamma_A$ is a fixed point of $c^{(1)}$, hence of c and c' (as $c'(B) \subseteq c(c'(B)) = B$ for a fixed point B) and of $\bar{c}$; it commutes with pullback because each $c^{(\alpha)}$ does and pullback preserves unions of chains; and it preserves binary intersections because each $c^{(\alpha)}$ does and, by (H1), $\bigcup_\alpha c^{(\alpha)}(A) \cap \bigcup_\beta c^{(\beta)}(B) = \bigcup_{\alpha,\beta} c^{(\alpha)}(A) \cap c^{(\beta)}(B) = \bigcup_\gamma c^{(\gamma)}(A \cap B)$. It lies above c and c', and if d is a closure operator above c and c' then $c^{(\alpha)}(A) \subseteq d(A)$ for all α by induction, d being monotone and idempotent, with unions at limits. □

A map in U_k is $\bar{c}$-bidense, since $c^{(k)} \leq \bar{c}$; so $V \subseteq U \subseteq W_{\bar{c}}$. The converse inclusion $W_{\bar{c}} \subseteq U$ is what the criterion is about, and its failure is detected by *witnesses*.

Definition 4.2. A *witness* for a subobject $S \subseteq Y$ is a regular epimorphism $f_S \colon X_S \to Y$ such that, for every word w and for $w = \bar{c}$, $\ker f_S \subseteq w(\Delta_{X_S})$ iff $w(S) = Y$. A *sum* of a sequence of maps $f_k \colon X_k \to Y_k$ is a map $F \colon X \to Y$ with monomorphisms $\sigma_k \colon X_k \hookrightarrow X$, $\tau_k \colon Y_k \hookrightarrow Y$, $F\sigma_k = \tau_k f_k$, such that $(\sigma_k \times \sigma_k)^{-1}(\ker F) = \ker f_k$ and $\tau_k^{-1}(\mathrm{im}F) = \mathrm{im}f_k$ for every k, and such that F is $\bar{c}$-bidense whenever every f_k is.

Theorem 4.3. *Assume (H1), that every subobject has a witness, and that every sequence of witnesses has a sum. Then $W_c \vee W_{c'} = W_{\bar{c}}$ iff $c^{(k)} = \bar{c}$ for some k.*

Proof. If $c^{(k)} = \bar{c}$, a $\bar{c}$-bidense map lies in $U_k \subseteq V$, and $V \subseteq W_{\bar{c}}$ always. If $c^{(k)} \neq \bar{c}$ for every k, choose for each k a subobject $S_k \subseteq Y_k$ with $c^{(k)}(S_k) \neq \bar{c}(S_k)$; replacing Y_k by $\bar{c}(S_k)$ we may assume that S_k is $\bar{c}$-dense in Y_k but $c^{(k)}(S_k) \neq Y_k$, as $c^{(k)} \leq \bar{c}$ and both commute with pullback along $\bar{c}(S_k) \hookrightarrow Y_k$. Let f_k be a witness for S_k and F a sum of the f_k. Each f_k is $\bar{c}$-bidense, being a regular epimorphism with $\ker f_k \subseteq \bar{c}(\Delta)$, so $F \in W_{\bar{c}}$. But $F \notin U_k$ for every k: pulling back along σ_k and τ_k, the conditions $c^{(k)}(\mathrm{im}F) = Y$ and $\ker F \subseteq c^{(k)}(\Delta_X)$ restrict, by pullback stability of $c^{(k)}$ and the definition of a sum, to $c^{(k)}(\mathrm{im}f_k) = Y_k$ and $\ker f_k \subseteq c^{(k)}(\Delta_{X_k})$, i.e. to $f_k \in U_k$, which fails since $c^{(k)}(S_k) \neq Y_k$. So $F \in W_{\bar{c}} \setminus U = W_{\bar{c}} \setminus V$ by Theorem 1.1. □

It remains to produce witnesses and sums. The witness is always the *fold map* $f_S \colon Y \amalg_S Y \to Y$ of the pushout of $Y \hookleftarrow S \hookrightarrow Y$; what has to be checked is the description of its kernel pair, and this is where the two classes of categories differ.

The case $k = 1$ has a formal description, which does not need witnesses.

Lemma 4.4. *For closure operators c, c' on an exact category, the composite cc' is a closure operator, and then $cc' = \bar{c}$, iff $c'c \leq cc'$. In that case $W_c \vee W_{c'} = W_{\bar{c}}$ by Theorem 1.1, since $U_1 = W_{cc'}$. If c and c' commute then $cc' = c'c = \bar{c}$.*

Proof. cc' is inflationary, monotone, pullback-stable and preserves intersections as a composite of such operators, so it is a closure operator iff it is idempotent. If $c'c \leq cc'$ then $cc'cc' \leq cccc'c' = cc'$; conversely, if cc' is idempotent then $c'c \leq cc'cc' = cc'$. A closure operator above c and c' is above cc', so $cc' = \bar{c}$. □

Coherent categories with effective unions. Assume

(T) finite unions of subobjects exist, are stable under pullback and are *effective*: $A \cup B$ is the pushout of $A \hookleftarrow A \cap B \hookrightarrow B$; pushouts of monomorphisms along monomorphisms exist, are pullback squares and have monic injections; and countable coproducts exist and are disjoint and universal.

Every Grothendieck topos satisfies (T) [4, A1.4, A2.4, C2.2], and (H1) as well, subobject lattices being frames. The category of groups does not: pushouts of monomorphisms are amalgamated free products, which are pullbacks with monic injections, but the union of the two factors in $\mathbb{Z} * \mathbb{Z}$ is not the whole group, and coproducts are not universal.

Proposition 4.5. *Assume (T). For $S \subseteq Y$ the fold map $f_S \colon X_S := Y \amalg_S Y \to Y$ is a witness, and the coproduct $\coprod_k f_k$ of a sequence of witnesses is a sum.*

Proof. Let $\iota_1, \iota_2 \colon Y \hookrightarrow X_S$ be the injections, monomorphisms with intersection S, which is also the equaliser of ι_1, ι_2. The union $\iota_1 Y \cup \iota_2 Y \subseteq X_S$ is the pushout of $\iota_1 Y \hookleftarrow S \hookrightarrow \iota_2 Y$ by effectiveness, hence the inclusion $\iota_1 Y \cup \iota_2 Y \hookrightarrow X_S$ is an isomorphism: the two injections *cover* X_S. The map f_S is a regular epimorphism, being split by ι_1. Since unions are pullback-stable, subobject lattices are distributive and $X_S \times X_S = \bigcup_{a,b} \iota_a Y \times \iota_b Y$; hence $\ker f_S = \bigcup_{a,b} \ker f_S \cap (\iota_a Y \times \iota_b Y)$, and $\ker f_S \cap (\iota_a Y \times \iota_b Y) = (\iota_a \times \iota_b)\big((\iota_a \times \iota_b)^{-1} \ker f_S\big) = (\iota_a \times \iota_b)(\Delta_Y)$, as $f_S \iota_a = f_S \iota_b = \mathrm{id}_Y$. For $a = b$ this is contained in Δ_{X_S}; for $a \neq b$ it is the image of (ι_1, ι_2) or (ι_2, ι_1). Now $(\iota_1, \iota_2)^{-1}(\Delta_{X_S}) = S$, so by pullback stability $(\iota_1, \iota_2)^{-1}(w(\Delta_{X_S})) = w(S)$, and the image of (ι_1, ι_2) lies in $w(\Delta_{X_S})$ iff $w(S) = Y$; the same holds for (ι_2, ι_1) by symmetry. So $\ker f_S \subseteq w(\Delta_{X_S})$ iff $w(S) = Y$.

For the sum, let $F = \coprod_k f_k \colon \coprod X_k \to \coprod Y_k$ with the coproduct injections σ_k, τ_k, monomorphisms since coproducts are disjoint. Since coproducts are universal, $(\coprod X_k)^2 = \coprod_{k,l} X_k \times X_l$, the pullback of $\ker F$ along $\sigma_k \times \sigma_k$ is $\ker f_k$ and the pullback of $\mathrm{im} F$ along τ_k is $\mathrm{im} f_k$ (images are stable under pullback). If every f_k is $\bar{c}$-bidense then F is a regular epimorphism, and $\ker F$ is the union of the images of the $\ker f_k$ (countable unions being images of coproducts), which lie in the images of $\bar{c}(\Delta_{X_k})$, hence in $\bar{c}(\Delta_{\coprod X_k})$ by Lemma 2.2(i), as $\Delta_{\coprod X_k}$ is the union of the images of the Δ_{X_k}. □

Abelian categories. Let $\mathcal{A}$ be a Grothendieck abelian category; it is exact, satisfies (H1) (subobject lattices are sets, unions of chains are directed colimits, exact and pullback-stable by AB5, and distribute over finite intersections), and has countable coproducts, but these are not disjoint and pushouts of monomorphisms are not unions in the sense of (T). For a map $f \colon X \to Y$ write $d \colon X \times X \to X$ for $(x, x') \mapsto x - x'$, a split epimorphism with $d^{-1}(0) = \Delta_X$ and $d^{-1}(K) = \ker f$ for the kernel object $K = \mathrm{Ker}\, f \subseteq X$. For a word w, pullback stability gives $w(\Delta_X) = d^{-1}(w(0))$, and since d is a split epimorphism, $d^{-1}(K) \subseteq d^{-1}(w(0))$ iff $K \subseteq w(0)$; so

$$(1) \quad \ker f \subseteq w(\Delta_X) \quad \text{iff} \quad \mathrm{Ker}\, f \subseteq w(0 \subseteq X) \quad \text{iff} \quad w(0 \subseteq \mathrm{Ker}\, f) = \mathrm{Ker}\, f,$$

the last step by pullback stability along $\mathrm{Ker}\, f \hookrightarrow X$. Call an object M *w-torsion* if $w(0 \subseteq M) = M$; by pullback stability along a quotient map, $w(S) = Y$ iff Y/S is w-torsion.

Proposition 4.6. *In a Grothendieck abelian category, for $S \subseteq Y$ the fold map $f_S \colon Y \amalg_S Y \to Y$ is a witness, and the direct sum $\bigoplus_k f_k$ of a sequence of witnesses is a sum.*

Proof. $X_S = (Y \oplus Y)/\{(s,-s) : s \in S\}$ and $f_S[(y_1,y_2)] = y_1 + y_2$ is an epimorphism with kernel object $\{[(y,-y)]\} \cong Y/S$. By (1), $\ker f_S \subseteq w(\Delta_{X_S})$ iff Y/S is w-torsion iff $w(S) = Y$. For the sum, $F = \bigoplus f_k$ has $\operatorname{Ker} F = \bigoplus \operatorname{Ker} f_k$ and $\operatorname{Coker} F = \bigoplus \operatorname{Coker} f_k$; the pullbacks of $\ker F$ and $\operatorname{im} F$ along the summand inclusions are $\ker f_k$ and $\operatorname{im} f_k$ by (1) and the corresponding statement for images. If every f_k is $\bar{c}$-bidense, then F is an epimorphism and $\operatorname{Ker} F = \bigoplus \operatorname{Ker} f_k$ is $\bar{c}$-torsion, because $\bar{c}(0 \subseteq \bigoplus K_k) \supseteq \bigoplus \bar{c}(0 \subseteq K_k)$ by Lemma 2.2(i) applied to the summand inclusions and the fact that unions of subobjects are sums; so $F \in W_{\bar{c}}$ by (1). □

Proof of Theorem 1.2. Theorem 4.3 with Lemma 4.1 and Proposition 4.5, respectively 4.6. □

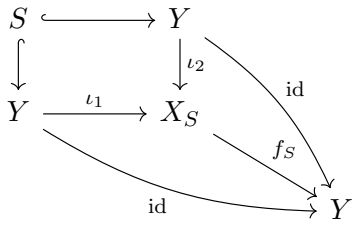


FIGURE 2. The witness: two copies of Y glued along S; the kernel pair of the fold map lies in $w(\Delta)$ exactly when $w(S) = Y$. In a topos the two copies cover X_S; in an abelian category the kernel of the fold map is Y/S.

Remark 4.7. Theorem 1.1 needs only exactness, and Proposition 3.5 only finite limits. In Theorem 4.3, (H1) serves to identify $\bar{c}$ with the transfinite iterate, witnesses to detect a single failure and sums to combine infinitely many; without sums one still has, for each fixed k, that $W_{\bar{c}} \subseteq U_k$ iff $c^{(k)} = \bar{c}$, provided $\bar{c}$ exists and witnesses do.

5. Instances

5.1. **Toposes.** Let $\mathcal{E}$ be a Grothendieck topos. Closure operators on $\mathcal{E}$ are the universal closure operators, in bijection with Lawvere–Tierney topologies $j \colon \Omega \to \Omega$ (the closure of $A \subseteq X$ is classified by $j \circ \chi_A$) and with subtoposes, and W_{c_j} is the class of maps inverted by the sheafification a_j [13, V.1], [4, A4.3]. The hypotheses (H1) and (T) hold, and the identity $c^{(k)} = \bar{c}$ on all subobject lattices is the identity $(jj')^k = j \vee j'$ on Ω (apply both sides to the generic subobject). Theorems 1.1 and 1.2 specialise to [23, Thms. 1.1, 1.2]:

Corollary 5.1. *For Lawvere–Tierney topologies j, j' on a Grothendieck topos, $W_j \vee W_{j'}$ is the class of maps whose image and kernel pair are reached by a bounded number of alternating closures, the closure of $W_j \cup W_{j'}$ under composition, and $W_j \vee W_{j'} = W_{j\vee j'}$ iff $(jj')^k = j \vee j'$ for some k.*

For a presheaf topos $\mathrm{Presh}(S)$ and Grothendieck topologies J, J' on the small category S the conditions are conditions on sieves: a map f is in U_k iff its image sieves $\{b : y \cdot b \in \mathrm{im} f\}$ and its equaliser sieves $\{b : x \cdot b = x' \cdot b\}$, $f(x) = f(x')$, are sent to the maximal sieve by $(jj')^k$. For topologies of ideal type, J_D and J_E for idempotent two-sided ideals D, E of S, this is local bijectivity along the ideal $(DE)^k$, and the criterion is the stabilisation of the chain $DE \supseteq DEDE \supseteq \cdots$, which holds on every finite category and fails on $(\mathbb{N}, \leq)$ for the ideals of the even and of the odd points [23, §5]. For ideals generated by two sets of objects A, B, the join of "bijective at the objects of A" and "bijective at the objects of B" is "bijective at $A \cap B$" iff every morphism through a long enough alternating zigzag of B and A factors through $A \cap B$; on $\mathbb{N}$ with the even and the odd points the join is the class of maps of towers bijective up to a uniform shift, a proper subclass of the isomorphisms of pro-sets [23, §6].

Remark 5.2. The classical sufficient condition for $W_j \vee W_{j'} = W_{j\vee j'}$ is that the alternating tower of sheafifications $(a_{j'}a_j)^k X$ reach the $(j \vee j')$-sheaves for one k and all X [15, Thm. 4.1]. The criterion of Theorem 1.2 is the iteration of the closure operators, and the two differ: on the contracted monoid of order 8 of [17, §7] the chain stabilises, so the join is preserved, while the strictly alternating plus tower never reaches the sheaves uniformly [17, Prop. 7.1], and a computation on the right M-sets with at most three elements shows that no composite of at most eight plus constructions, in particular no $(a_E a_D)^k$ with $k \leq 2$, is a sheaf for all of them [23, §8].

The topos-theoretic counterpart of the stable torsion theories of Theorem 1.4 is given by the open, the closed and the dense topologies, which satisfy the criterion with $k = 1$ against *every* topology. For a subobject A of X write $b(A) := \{B \Rightarrow A : B \in \mathrm{Sub}(X)\}$, the smallest sublocale of the Heyting algebra $\mathrm{Sub}(X)$ containing A; the fixed points of any nucleus on $\mathrm{Sub}(X)$ form a sublocale. The two-step factorisations below all arise from the following lemma.

Lemma 5.3. *Let c, c' be universal closure operators on an elementary topos with $c'c \leq cc'$, so that $cc' = \bar{c}$ by Lemma 4.4; let a' be the sheafification for c', with units ℓ, and let $f\colon X \to Y$ lie in $W_{\bar{c}}$. Then the kernel pair of $a'(f)$ lies in $c(\Delta_{a'X})$. Consequently, if the image of $a'(f)$ is c-dense, then $a'(f) \in W_c$, and f is the composite of the induced map $g\colon X \to Z := Y \times_{a'Y} a'X$, which lies in $W_{c'}$, and the projection $h\colon Z \to Y$, which lies in W_c; so $W_{\bar{c}} = W_c \circ W_{c'}$ as soon as the image of $a'(f)$ is c-dense for every $f \in W_{\bar{c}}$.*

Proof. Put $F = a'X$, $G = a'Y$, $\varphi = a'(f)$ and $\ell = \ell_X$. Since F is c'-separated, Δ_F is c'-closed, and $\ker \ell = c'(\Delta_X)$: "$\subseteq$" because $\ell \in W_{c'}$, and

"$\supseteq$" because $(\ell \times \ell)(c'(\Delta_X)) \subseteq c'((\ell \times \ell)(\Delta_X)) \subseteq c'(\Delta_F) = \Delta_F$. Hence

$$\begin{aligned}(\ell \times \ell)^{-1}(\ker \varphi) = \ker(\ell_Y f) &= (f \times f)^{-1}(\ker \ell_Y) \subseteq (f \times f)^{-1} c'(\Delta_Y) \\ &= c'(\ker f) \subseteq c'cc'(\Delta_X) \subseteq cc'(\Delta_X) \\ &= c(\ker \ell) = (\ell \times \ell)^{-1} c(\Delta_F).\end{aligned}$$

The image I of $\ell \times \ell$ is $\mathrm{im}\ell \times \mathrm{im}\ell$, which is c'-dense in $F \times F$, and $\ker\varphi \cap I = (\ell \times \ell)((\ell \times \ell)^{-1} \ker \varphi) \subseteq (\ell \times \ell)((\ell \times \ell)^{-1} c(\Delta_F)) \subseteq c(\Delta_F)$; so $\ker\varphi \subseteq c'(\ker\varphi) \cap c'(I) = c'(\ker\varphi \cap I) \subseteq c'c(\Delta_F) \subseteq cc'(\Delta_F) = c(\Delta_F)$. For the factorisation, h is a pullback of $\varphi \in W_c$ and W_c is stable under pullback (dense subobjects and closures are), so $h \in W_c$; the other projection $p\colon Z \to a'X$ is a pullback of $\ell_Y \in W_{c'}$, so $p \in W_{c'}$, and $pg = \ell_X \in W_{c'}$, so $g \in W_{c'}$ by two-out-of-three. The reverse inclusion $W_c \circ W_{c'} \subseteq W_{\bar{c}}$ is clear. □

Proposition 5.4. *Let $\mathcal{E}$ be an elementary topos and c, c' universal closure operators.*

(i) *If $c(A) \in b(A)$ for every subobject A of every X, i.e. $(c(A) \Rightarrow A) \Rightarrow A = c(A)$, then $cc' = c \vee c'$ for every c', hence $W_c \vee W_{c'} = W_{c \vee c'}$. This holds for the open topology $o(U)$ of a subterminal U, with $c(A) = U_X \Rightarrow A$ where $U_X = U \times X$, and for the double-negation topology, with $c(A) = \neg\neg A = \neg A \Rightarrow A$.*

(ii) *If c is the closed topology $c(U)$, $c(A) = A \cup U_X$, then $c'c = c \vee c'$ for every c', hence $W_c \vee W_{c'} = W_{c \vee c'}$.*

(iii) *For the open topology $o = o(U)$ and any topology j, $W_{o \vee j} = W_o \circ W_j$: a map $f\colon X \to Y$ inverted by the sheafification for $o \vee j$ is the composite of the induced map $X \to Y \times_{a_j Y} a_j X$, which is j-bidense, and the projection to Y, which is o-bidense.*

(iv) *For the closed topology $c = c(U)$ and any topology j, $W_{c \vee j} = W_j \circ W_c$: a map $f\colon X \to Y$ inverted by the sheafification for $c \vee j$ is the composite of the induced map $X \to Y \times_{a_c Y} a_c X$, which is c-bidense, and the projection to Y, which is j-bidense.*

(v) *For the double-negation topology and any topology j, $W_{\neg\neg \vee j} = W_{\neg\neg} \circ W_j$, with the middle term $Y \times_{a_j Y} a_j X$ of (iii).*

Proof. (i) Let $B \subseteq X$ and $A := c'(B)$. The fixed points of c'_X form a sublocale of $\mathrm{Sub}(X)$ containing A, hence containing $b(A) \ni c(A)$; so $c'(cc'(B)) = cc'(B)$ and $cc'cc' = ccc' = cc'$, and Lemma 4.4 applies. $U_X \Rightarrow A$ lies in $b(A)$ by definition, and $\neg\neg A = \neg A \Rightarrow A$ in any Heyting algebra. (ii) $c'(A \cup U_X) \supseteq c'(A) \cup U_X$, so $cc' \leq c'c$ and Lemma 4.4 applies with the roles exchanged. (iii) Write j_U for the closure operator on the topos $\mathcal{E}/U$ that closes a subobject A of an object $X \to U$ as j does in $\mathcal{E}$; since $\mathrm{Sub}_{\mathcal{E}/U}(X \to U) = \mathrm{Sub}_{\mathcal{E}}(X)$ and pullbacks in $\mathcal{E}/U$ are computed in $\mathcal{E}$, it is a universal closure operator, i.e. a topology on $\mathcal{E}/U$, and a map over U is j_U-bidense iff it is j-bidense in $\mathcal{E}$. We use three facts. (1) $q \in W_o$ iff $q \times U$ is an isomorphism: a subobject $A \subseteq X$ is o-dense iff $U_X \Rightarrow A = X$ iff $U_X \leq A$ iff $A \times U = X \times U$, and $(-) \times U$, the pullback along $U \to 1$, preserves images

and kernel pairs; so the image of q is dense iff $q \times U$ is epimorphic, and the kernel pair $K(q)$ lies in $U_{X\times X} \Rightarrow \Delta_X$ iff $K(q) \times U \leq \Delta_X \times U$ iff $q \times U$ is monic. (2) $f \in W_{o\vee j}$ iff $f \times U$ is j_U-bidense: by (i) the closure operator of $o \vee j$ is $c_o c_j$, and $c_o c_j(\mathrm{im} f) = Y$ iff $U_Y \leq c_j(\mathrm{im} f)$ iff $c_j(\mathrm{im} f) \times U = Y \times U$ iff $\mathrm{im}(f \times U) = \mathrm{im} f \times U$ is j-dense in $Y \times U$, the closure commuting with the pullback along $Y \times U \to Y$; likewise $K(f) \leq U_{X\times X} \Rightarrow c_j(\Delta_X)$ iff $K(f) \times U \leq c_j(\Delta_X) \times U = c_j(\Delta_X \times U)$, and $K(f) \times U$ and $\Delta_X \times U$ are the kernel pair of $f \times U$ and the diagonal of $X \times U$ in $\mathcal{E}/U$. (3) $a_{j_U}(X \times U \to U) \cong (a_j X \times U \to U)$, naturally in X: for a j-sheaf F the object $F \times U \to U$ is a j_U-sheaf, because a map $A \to F \times U$ over U from a j-dense subobject $A \subseteq B$ of an object $B \to U$ is a map $A \to F$, which extends uniquely to $B \to F$ and hence to $B \to F \times U$ over U; and $\ell_X \times U \colon X \times U \to a_j X \times U$ is j_U-bidense as a pullback of the unit $\ell_X \in W_j$, hence is inverted by a_{j_U}. Now let $f \in W_{o\vee j}$. By (2) and (3), $a_j(f) \times U \cong a_{j_U}(f \times U)$ is an isomorphism, so $a_j(f) \in W_o$ by (1), and Lemma 5.3 (with $c = c_o$, $c' = c_j$, for which $c'c \leq cc'$ by (i)) gives the factorisation. Conceptually, (3) says that the open subtopos $\mathcal{E}/U \hookrightarrow \mathcal{E}$ pulls back along the inclusion $i \colon \mathrm{Sh}_j(\mathcal{E}) \hookrightarrow \mathcal{E}$ to $\mathrm{Sh}_j(\mathcal{E})/i^*U \simeq \mathrm{Sh}_{j_U}(\mathcal{E}/U)$, cf. [4, A4.5]; the argument above does not depend on this.

(iv) The c-sheaves are the objects F with $F \times U \cong U$, and $a_c X$ is the pushout of $X \hookleftarrow X \times U \to U$ [4, A4.5]: slicing over U preserves pushouts and $U \times U = U$, so the pushout is trivial over U, and a map $X \to F$ with F trivial over U restricts to $X \times U$ as the composite $X \times U \to U \to F$, the only map $X \times U \to F$, since $\mathrm{Hom}(X \times U, F) = \mathrm{Hom}_{/U}(X \times U, F \times U) = \mathrm{Hom}_{/U}(X \times U, U)$. Let $q = \ell_X \colon X \to a_c X$ and $u \colon U \to a_c X$ be the coprojections, which are jointly epimorphic, q', u' those of Y; then $a_c(f)q = q'f$ and $a_c(f)u = u'$. By (ii) the closure operator of $c \vee j$ is $c_j c_c$, so $f \in W_{c\vee j}$ means $c_j(\mathrm{im} f \cup U_Y) = Y$ and $\ker f \subseteq c_j(\Delta_X \cup U_{X\times X})$. We show that $a_c(f) \in W_j$; the factorisation then follows from Lemma 5.3, applied to the pair (c_j, c_c), for which $c_c c_j \leq c_j c_c$ by (ii). Recall that for a jointly epimorphic family (e_i) with codomain B and subobjects A, A' of B, $A \subseteq A'$ iff $e_i^{-1}(A) \subseteq e_i^{-1}(A')$ for all i, as $A = \bigcup_i e_i(e_i^{-1}(A))$. Image: let $B = \mathrm{im} a_c(f)$. Then $q'^{-1}(B) \supseteq \mathrm{im} f$, since $a_c(f)q = q'f$, and $q'^{-1}(B) \supseteq U_Y$, since q' restricted to $U_Y = Y \times U$ factors through u' and $u'(U) = a_c(f)(u(U)) \subseteq B$; so $q'^{-1}(c_j B) = c_j(q'^{-1}B) \supseteq c_j(\mathrm{im} f \cup U_Y) = Y$, and u' factors through $B \subseteq c_j B$; hence $c_j B = a_c Y$. Kernel pair: the four maps $q \times q$, $q \times u$, $u \times q$, $u \times u$ are jointly epimorphic onto $a_c X \times a_c X$, and it suffices to check $\ker a_c(f) \subseteq c_j(\Delta)$ after pulling back along each, where the pullback of $c_j(\Delta)$ is the j-closure of the pullback of Δ. Along $q \times u$ and $u \times q$ the pullback of Δ is all of $X \times U$, because $qp_X = up_U$ on $X \times U$; along $u \times u$ the domain $U \times U = U$ maps into Δ. Along $q \times q$: $(q \times q)^{-1} \ker a_c(f) = (f \times f)^{-1} \ker q'$,

and $\ker q' \subseteq c_c(\Delta_Y) = \Delta_Y \cup U_{Y\times Y}$ because $q' = \ell_Y$ is c-bidense; so

$$\begin{aligned}(q\times q)^{-1}\ker a_c(f) &\subseteq \ker f \cup U_{X\times X} \subseteq c_j(\Delta_X \cup U_{X\times X})\\ &\subseteq c_j(\ker q) = (q\times q)^{-1}c_j(\Delta_{a_cX}),\end{aligned}$$

using $\Delta_X \cup U_{X\times X} \subseteq \ker q$, which holds because q restricted to U_X factors through u. Hence $a_c(f) \in W_j$.

(v) By (i) the closure operator of $\neg\neg \vee j$ is $\neg\neg c_j$ and $c_j\neg\neg \leq \neg\neg c_j$, so by Lemma 5.3 it suffices to show that for $f \in W_{\neg\neg\vee j}$ the image B of $\varphi := a_j(f)\colon F := a_jX \to G := a_jY$ is $\neg\neg$-dense, i.e. $\neg B = 0$. First, $\neg\neg c_j(B) = G$: from $\varphi\ell_X = \ell_Y f$ we get $\mathrm{im}f \subseteq \ell_Y^{-1}(B)$, so $Y = \neg\neg c_j(\mathrm{im}f) \subseteq \neg\neg c_j(\ell_Y^{-1}B) = \ell_Y^{-1}(\neg\neg c_jB)$, i.e. $\mathrm{im}\ell_Y \subseteq \neg\neg c_jB$, and as $\mathrm{im}\ell_Y$ is j-dense, $G = c_j(\mathrm{im}\ell_Y) \subseteq c_j\neg\neg c_jB \subseteq \neg\neg c_j\neg\neg c_jB = \neg\neg c_jB$. Second, $c_j(0_G) \subseteq B$, where 0_G is the least subobject of G: the closed subobject $c_j(0_1)$ of the sheaf 1 is a sheaf receiving a dense monomorphism from 0, hence is the initial object $a_j(0)$ of $\mathrm{Sh}_j(\mathcal{E})$; $c_j(0_G) = G \times c_j(0_1)$ and $c_j(0_F) = F \times c_j(0_1)$ are then initial objects of $\mathrm{Sh}_j(\mathcal{E})$ too, since they are sheaves mapping to $a_j(0)$ and the initial object of the topos $\mathrm{Sh}_j(\mathcal{E})$ is strict; so φ, which maps $c_j(0_F)$ into $c_j(\varphi(0_F)) = c_j(0_G)$, restricts to a map between initial objects, an isomorphism, and $c_j(0_G) = \varphi(c_j(0_F)) \subseteq B$. Third, if $A \cap B = 0$ then $A \cap c_jB \subseteq c_jA \cap c_jB = c_j(A \cap B) = c_j(0_G) \subseteq B$, so $A \cap c_jB \subseteq A \cap B = 0$ and $A \subseteq \neg c_jB = 0$. Hence $\neg B = 0$. □

Proposition 5.5. *Let c, c' be universal closure operators on an elementary topos with $c'c \leq cc'$, and let a, a' be the sheafifications for c and c'.*

(a) *For every c'-sheaf G the c-sheaf aG is c'-separated.*
(b) *The following are equivalent: (1) $a'(f) \in W_c$ for every $f \in W_{\bar{c}}$; (1′) $a'(f) \in W_c$ for every $f \in W_c$, i.e. a' preserves c-bidense maps; (2) aG is a c'-sheaf for every c'-sheaf G; (3) the unit $G \to \bar{a}G$ induces an isomorphism $aG \cong \bar{a}G$ for every c'-sheaf G, where $\bar{a}$ is the sheafification for $\bar{c} = cc'$, i.e. $aa' \cong \bar{a}$ naturally, compatibly with the units. They imply $W_{\bar{c}} = W_c \circ W_{c'}$.*
(c) *If a c'-sheaf G is c-separated, then $aG \cong \bar{a}G$ is a c'-sheaf.*

Proof. (a) Let $\ell\colon G \to aG$ be the unit; $\ker\ell = c(\Delta_G)$ as in the proof of Lemma 5.3, and $c'(\Delta_G) = \Delta_G$ as G is c'-separated. Hence $(\ell\times\ell)^{-1}c'(\Delta_{aG}) = c'(\ker\ell) = c'c(\Delta_G) \subseteq cc'(\Delta_G) = c(\Delta_G) = (\ell\times\ell)^{-1}(\Delta_{aG})$. With $I = \mathrm{im}(\ell\times\ell)$, which is c-dense in $aG \times aG$, this gives $c'(\Delta_{aG}) \cap I \subseteq \Delta_{aG}$, so $c'(\Delta_{aG}) \subseteq c(c'(\Delta_{aG}) \cap I) \subseteq c(\Delta_{aG}) = \Delta_{aG}$, using $c(S) = c(S) \cap c(I) = c(S \cap I)$ and the c-separatedness of aG.

(b) (1)⇒(1′) as $W_c \subseteq W_{\bar{c}}$. (1′)⇒(1): by Lemma 4.4 and Theorem 1.1 with $k = 1$, a map $f \in W_{\bar{c}} = U_1$ is a composite $m_2m_1e_2e_1$ with $e_1, m_1 \in W_{c'}$ and $e_2, m_2 \in W_c$; a' inverts e_1 and m_1, and $a'(e_2), a'(m_2) \in W_c$ by (1′), so $a'(f) \in W_c$. (2)⇔(3): a c-sheaf that is a c'-sheaf is a $\bar{c}$-sheaf, since $\mathrm{Sh}_{\bar{c}} = \mathrm{Sh}_c \cap \mathrm{Sh}_{c'}$, and the unit $G \to aG$ is c-bidense, hence $\bar{c}$-bidense, so $aG \cong \bar{a}G$; conversely $\bar{a}G$ is a c'-sheaf. (3)⇒(1): for $f \in W_{\bar{c}}$ put $\varphi = a'(f)\colon F \to G$, a

map of c'-sheaves. By (3), $a(\varphi) \cong \bar{a}(\varphi) \cong \bar{a}(f)$, the last because $\bar{a}$ inverts the units of a', which lie in $W_{c'} \subseteq W_{\bar{c}}$; and $\bar{a}(f)$ is an isomorphism. So $a(\varphi)$ is an isomorphism, i.e. $\varphi \in W_c$. (1)⇒(2): let G be a c'-sheaf and $f\colon G \to \bar{a}G$ the unit, which lies in $W_{\bar{c}}$; as G and $\bar{a}G$ are c'-sheaves, $a'(f) \cong f$, so $f \in W_c$ by (1). Then $a(f)$ is an isomorphism, and so is $a(\bar{a}G) \to \bar{a}G$, as $\bar{a}G$ is a c-sheaf; the composite $aG \xrightarrow{a(f)} a(\bar{a}G) \xrightarrow{\cong} \bar{a}G$ is the map induced by the unit, so $aG \cong \bar{a}G$ is a c'-sheaf. Finally (1) gives $W_{\bar{c}} = W_c \circ W_{c'}$ by Lemma 5.3.

(c) Let $f\colon G \to S := \bar{a}G$ be the unit and $G_0 = \mathrm{im} f$. Its kernel pair is $\bar{c}(\Delta_G) = cc'(\Delta_G) = c(\Delta_G) = \Delta_G$, so f is a monomorphism, and $cc'(G_0) = S$ as f is $\bar{c}$-bidense. Now S is a c'-sheaf, being a $\bar{c}$-sheaf, and $G_0 \cong G$ is a c'-sheaf; a subobject of a c'-sheaf is a c'-sheaf iff it is c'-closed [13, V.3], [4, A4.3] (the closure $c'(G_0)$ is a c'-sheaf and $G_0 \hookrightarrow c'(G_0)$ is a c'-dense monomorphism of c'-sheaves, hence an isomorphism). So $c'(G_0) = G_0$ and $c(G_0) = cc'(G_0) = S$: f is a c-dense monomorphism into a c-sheaf, i.e. the c-sheafification of G. □

Thus Proposition 5.4(iii)–(v) say that the open, the closed and the double-negation sheafifications are preserved by, or preserve, every other sheafification in the sense of (2), and (c) says that (2) can only fail for c'-sheaves that are not c-separated; Lemma 5.7 makes this precise, and Example 5.8 shows that (2) does fail in general, even for commuting c, c'. What the c'-sheafification always preserves under $c'c \leq cc'$ is the c-dense monomorphisms.

Proposition 5.6. *Let c, c' be universal closure operators on an elementary topos with $c'c \leq cc'$, and let a' be the sheafification for c', with unit η. Then $\eta^{-1}(a'(I)) = c'(I)$ for every subobject I of every Y, and a' preserves c-dense monomorphisms: if $I \subseteq Y$ is c-dense, then $a'(I)$ is c-dense in $a'(Y)$.*

Proof. Let $\eta\colon Y \to a'Y$ be the unit. We first check that $\eta^{-1}(a'(I)) = c'(I)$ for every subobject I of Y. The pullback contains I, by naturality of η, and is c'-closed, as $a'(I)$ is a c'-sheaf, hence a c'-closed subobject of the c'-sheaf $a'Y$ [13, V.3], [4, A4.3], and closed subobjects pull back to closed subobjects. Conversely let $y\colon U \to Y$ have $\eta y \in a'(I)$. The image of the unit $I \to a'(I)$ is c'-dense, so there are a c'-dense $V \subseteq U$ and an epimorphism $p\colon V' \to V$ with $\eta y p = \eta x$ for some $x\colon V' \to I \subseteq Y$; then (yp, x) lies in $\ker \eta = c'(\Delta_Y)$ (proof of Lemma 5.3), so yp and x agree on a c'-dense subobject of V', on which yp factors through I; hence yp factors through $c'(I)$, so does $y|_V$ as p is epimorphic, and $y^{-1}(c'(I)) = c'(y^{-1}(I))$ is closed and contains the dense V, so $y \in c'(I)$.

Now let I be c-dense and put $T := c(a'(I)) \subseteq a'Y$. Since $a'(I)$ is c'-closed, $c'(T) = c'cc'(a'(I)) \subseteq cc'c'(a'(I)) = T$: T is c'-closed. By universality, $\eta^{-1}(T) = c(\eta^{-1}(a'(I))) = cc'(I) \supseteq c(I) = Y$, so the image of η lies in T; it is c'-dense, so $a'Y = c'(\mathrm{im}\eta) \subseteq T$, i.e. $a'(I)$ is c-dense. □

So under $c'c \leq cc'$ the c'-sheafification preserves the monomorphisms in W_c, and by Proposition 5.5(b) the two-step factorisation $W_{\bar{c}} = W_c \circ W_{c'}$

holds iff it also preserves the c-bidense epimorphisms. It need not, even when c and c' commute:

Lemma 5.7. *Let $c'c \leq cc'$ and let G be a c'-sheaf, with c-separated quotient $G^s = G/c(\Delta_G)$.*

(i) *G^s is c-separated and c'-separated, and $aG \cong a(G^s)$.*
(ii) *If H is c-separated and c'-separated, then $a'H$ is c-separated (for any c, c').*
(iii) *aG is a c'-sheaf iff G^s is c-dense in $a'(G^s)$.*

Proof. (i) Put $R = c(\Delta_G)$ and let $q\colon G \to G^s$ be the quotient, with kernel pair R. The unit $G \to aG$ has kernel pair R and factors through q, so q lies in W_c and $aG \cong a(G^s)$. As $q \times q$ is an epimorphism, a subobject of $G^s \times G^s$ is determined by its pullback along $q \times q$, and pullback commutes with c and c'; the pullback of Δ_{G^s} is R, so G^s is c-separated because $c(R) = R$, and c'-separated because $c'(R) = c'c(\Delta_G) \subseteq cc'(\Delta_G) = c(\Delta_G) = R$, using $c'(\Delta_G) = \Delta_G$.

(ii) The unit $\ell\colon H \to a'H$ is a c'-dense monomorphism. Let $x, y\colon U \to a'H$ agree on a c-dense $U_1 \subseteq U$. Then $U_2 = x^{-1}(H) \cap y^{-1}(H)$ is c'-dense in U and $x|_{U_2} = \ell x'$, $y|_{U_2} = \ell y'$; x', y' agree on $U_1 \cap U_2$, which is c-dense in U_2, hence on U_2 as H is c-separated. So x, y agree on the c'-dense U_2, hence $x = y$ as $a'H$ is c'-separated.

(iii) Put $H = a'(G^s)$; by (i) and (ii) H is a c-separated c'-sheaf and $G^s \hookrightarrow H$ is a monomorphism. By Proposition 5.5(c), $aH \cong \bar{a}H$ is a c'-sheaf, and $\bar{a}H \cong \bar{a}(G^s)$ as $\bar{a}$ inverts the units of a'. If G^s is c-dense in H, then $aG \cong a(G^s) \cong aH$ is a c'-sheaf. Conversely, if $aG \cong a(G^s)$ is a c'-sheaf, it is a $\bar{c}$-sheaf and the c-bidense unit $G^s \to a(G^s)$ is $\bar{c}$-bidense, so $a(G^s) \cong \bar{a}(G^s) \cong \bar{a}H$, and the unit $G^s \to \bar{a}H$ factors as $G^s \hookrightarrow H \to \bar{a}H$, where $H \to \bar{a}H$ is a monomorphism, its kernel pair being $cc'(\Delta_H) = c(\Delta_H) = \Delta_H$. A generalised element $y\colon U \to H$ composed with $H \to \bar{a}H$ lies in $c(G^s) = \bar{a}H$, so on a c-dense $U_1 \subseteq U$ it factors through $G^s \to \bar{a}H$; as $H \to \bar{a}H$ is monic, $y|_{U_1}$ factors through G^s, i.e. $y \in c(G^s)$. Hence G^s is c-dense in H. □

Example 5.8. Let $\mathcal{C}$ be the category with two objects a, b, $\mathcal{C}(a,a) = \{1, u, v\}$ with $gu = u$ and $gv = v$ for all g, $\mathcal{C}(a,b) = \{h\}$ with $hu = hv = h$, $\mathcal{C}(b,b) = \{1\}$ and $\mathcal{C}(b,a) = \emptyset$, and let $\mathcal{E}$ be the presheaf topos on $\mathcal{C}$. A presheaf X is a set $X(a)$ with a right action of the monoid $\{1, u, v\}$, a set $X(b)$ and a map $X(b) \to X(a)$, $y \mapsto yh$, whose image consists of common fixed points of u and v. Let c be the Grothendieck topology whose least covering sieves are $\{u, v\}$ on a and the maximal sieve on b, and c' the one with the maximal sieve on a and $\{h\}$ on b; these are sieves, stable under pullback and transitive, because $\{u, v\}$ is closed under composition on either side and $h\{1, u, v\} = \{h\}$. The closure operators are

$$c(A)(a) = \{x : xu, xv \in A(a)\}, \qquad c(A)(b) = A(b),$$
$$c'(A)(a) = A(a), \qquad c'(A)(b) = \{y : yh \in A(a)\};$$

they commute, $cc' = c'c = \bar{c}$, because $hu = hv = h$, and they are incomparable, because $\{u, v\}$ is not a c'-cover and $\{h\}$ is not a c-cover. A presheaf X is a c-sheaf iff $x \mapsto (xu, xv)$ is injective on $X(a)$, with no condition on $X(b)$: a matching family on $\{u, v\}$ is a pair (x_u, x_v) with $x_u u = x_u$, $x_u v = x_v$, $x_v u = x_u$, $x_v v = x_v$, and it is the restriction of x_u. It is a c'-sheaf iff $y \mapsto yh$ is a bijection from $X(b)$ onto the set of common fixed points of u and v in $X(a)$, with no condition on $X(a)$.

Let P be the presheaf with $P(a) = \{x_u, x_v, w\}$, $xu = x_u$ and $xv = x_v$ for all $x \in P(a)$, and $P(b) = \emptyset$. It is a c'-sheaf, as no element of $P(a)$ is fixed by both u and v. Its c-sheafification is the subterminal object V with $V(a) = 1$ and $V(b) = \emptyset$: the map $P \to V$ is c-bidense, all elements of $P(a)$ having the same images under u and v, and V is a c-sheaf. But V is not a c'-sheaf, its element at a being fixed while $V(b) = \emptyset$. So a does not preserve c'-sheaves, and (1)–(3) of Proposition 5.5(b) fail; explicitly, $\bar{a}P = 1$, as $P \to 1$ is $\bar{c}$-bidense (V is c'-dense in 1), while $aa'P = aP = V$.

The two-step factorisation fails as well: $f\colon P \to 1$ lies in $W_{\bar{c}}$ but not in $W_c \circ W_{c'}$. Suppose $f = qp$ with $p\colon P \to Z$ in $W_{c'}$ and $q\colon Z \to 1$ in W_c. Since the only c-cover of b is the maximal sieve, $c(S)(b) = S(b)$ for every subobject S of every presheaf; so $q \in W_c$ forces $Z(b)$ to be a singleton $\{z\}$, and $zh \in Z(a)$ is fixed by u and v. The c'-sheafification of Z is the presheaf Z' with $Z'(a) = Z(a)$ and $Z'(b)$ the set of common fixed points of u, v in $Z(a)$, the unit being the identity at a and $z \mapsto zh$ at b. As $p \in W_{c'}$, $a'(p)\colon a'P = P \to Z'$ is an isomorphism; but $P(b) = \emptyset$ and $zh \in Z'(b)$. In the other order f is the composite of the c-bidense map $P \to V$ and the c'-dense monomorphism $V \hookrightarrow 1$; in fact $W_{\bar{c}} = W_{c'} \circ W_c$, by Proposition 5.5(b) for the pair (c', c), since a' preserves c-sheaves, the c-sheaf condition concerning $X(a)$ only, which a' does not change. As in Example 5.19 the closure operators commute and the factorisation holds in exactly one order. In terms of Lemma 5.7(iii), the c-separated quotient $P^s = V$ of P is not c-dense in $a'(P^s) = 1$; and a' does not preserve the c-bidense epimorphism $P \to V$, since $a'(P \to V)$ is the map $P \to 1$, whose image V is not c-dense.

5.2. **Locales.** Let L be a frame and $\mathrm{Sh}(L)$ its topos of sheaves. Closure operators on $\mathrm{Sh}(L)$ correspond to the nuclei on L (sublocales), ordered as the nuclei [11, II.2], [4, A4.5]; the subobjects of the representable $y(U)$ are the $V \leq U$; and for the nucleus n with sublocale inclusion i, the sheafification is $i_* i^*$ with $i^*(y(V)) = y(n(V))$, so $V \hookrightarrow W$ is dense iff $n(V) = n(W)$, and the closure of V in $y(U)$ is $n(V) \wedge U$, the largest $W \leq U$ with $n(W) = n(V)$. By induction, using $U \leq n(U), n'(U)$, $(cc')^k(V) = (nn')^k(V) \wedge U$ and $\bar{c}(V) = (n \vee n')(V) \wedge U$ [23, Lemma 7.1].

Corollary 5.9. *For nuclei n, n' on a frame L, $W_n \vee W_{n'} = W_{n \vee n'}$ in* $\mathrm{Same}(\mathrm{Sh}(L))$ *iff $(nn')^k = n \vee n'$ in the frame of nuclei for some k.*

Proof. Theorem 1.2 and the formula above with $U = \top$. □

Open and closed nuclei satisfy the condition with $k = 1$: $o(U)c(U')$, $V \mapsto U \to (U' \vee V)$, is idempotent and hence the join. For a topological space X and subspaces A, B, with nuclei $n_A(V) = \mathrm{int}(V \cup (X \setminus A))$, the class W_{n_A} consists of the maps of sheaves whose inverse image to A is an isomorphism, and the condition reads: the alternating derivation $T(F) = \mathrm{cl}(\mathrm{cl}(F \cap B) \cap A)$ on closed sets reaches its transfinite limit in a uniform number of steps [23, Prop. 7.5]. It holds for closed and for open subspaces with $k = 1$ and fails for the rationals and the irrationals in $\mathbb{R}$: for a countable closed set C whose points of even Cantor–Bendixson rank are rational and of odd rank irrational, $T^j(C)$ is the $2j$-th derived set, nonempty for $2j$ below the rank, while the transfinite limit is empty; so in $\mathrm{Same}(\mathrm{Sh}(\mathbb{R}))$ the join of "isomorphism over $\mathbb{Q}$" and "isomorphism over the irrationals" is strictly smaller than the class of the meet subtopos, although the two sublocales meet in a nonempty dense sublocale [23, Ex. 7.6].

5.3. **Frames as categories: the pointwise criterion.** Let L be a frame, or just a meet-semilattice with top, viewed as a category. It is exact: every morphism $x \le y$ is a monomorphism, the only regular epimorphisms are the identities, and the (regular epi, mono) factorisation of $x \le y$ is $x \le y$ itself; kernel pairs are diagonals and equivalence relations trivial. $\mathrm{Sub}(y) = {\downarrow}y$, and the pullback of $z \le y$ along $x \le y$ is $z \wedge x$. For a nucleus n on L put $c_n(z \le y) := n(z) \wedge y$; this is a closure operator: inflationary, monotone, idempotent ($n(n(z) \wedge y) \wedge y = n(z) \wedge n(y) \wedge y = n(z) \wedge y$), intersection-preserving, and pullback-stable ($c_n(z \wedge x) = n(z) \wedge n(x) \wedge x = c_n(z) \wedge x$). The c_n-bidense maps are the $x \le y$ with $n(x) \wedge y = y$, i.e. $n(x) \ge y$, i.e. $n(x) = n(y)$: W_{c_n} is the class $W_n^L = \{x \le y : nx = ny\}$ of [21, Thm. 1.1].

Corollary 5.10. *For nuclei n, n' on L, in* $\mathrm{Same}(L)$*,*

$$W_n^L \vee W_{n'}^L = \{x \le y : \ (nn')^k x \ge y \text{ for some } k\},$$

every such $x \le y$ being the composite of the chain $x \le n'x \wedge y \le nn'x \wedge y \le \cdots \le y$ of inequalities alternately in $W_{n'}^L$ and W_n^L; and $W_n^L \vee W_{n'}^L = W_{n \vee n'}^L$ iff for every x some iterate $(nn')^k x$ equals $(n \vee n')x$.

Proof. Theorem 1.1: $x \le y$ is in U_k iff $c^{(k)}(x) = (nn')^k x \wedge y = y$, and the peeling factorisation is the displayed chain. For the last assertion, $W_{n \vee n'}^L = \{x \le y : (n \vee n')x \ge y\}$; if the pointwise identity holds, then $(n \vee n')x \ge y$ gives $(nn')^k x \ge y$ for the k of x; conversely take $y = (n \vee n')x$. □

On a frame the one-sided commutations of Proposition 5.4 characterise the nuclei involved. For $x \in L$ let $b(x) = \{y \to x : y \in L\}$ be the smallest sublocale containing x, with nucleus $r_x(w) = (w \to x) \to x$.

Proposition 5.11. *Let n be a nucleus on a frame L.*

(a) *$n \vee m = n \circ m$ for every nucleus m iff $n(S) \subseteq S$ for every sublocale S iff $n(x) \in b(x)$ for every x. Open nuclei and the double-negation nucleus have this property; if the filter $\{y : n(y) = 1\}$ has a least*

element, in particular if L is finite, the property holds only for open nuclei.

(b) *$n \vee m = m \circ n$ for every nucleus m iff n is closed.*

In particular $(u_a \vee m)(x) = a \to m(x)$ and $(c_b \vee m)(x) = m(b \vee x)$ for all nuclei m (these identities are standard, cf. [11, II.2]*), and $\neg\neg \vee m = \neg\neg \circ m$.*

Proof. (a) By Lemma 4.4 (with Sub replaced by L) $n \vee m = nm$ iff nm is idempotent. If $n(S) \subseteq S$ for all S then $n(m(x))$ is fixed by m and $nmnm = nnm = nm$. If nm is idempotent for all m, take $m = r_x$: $nr_xn(x) = nr_xnr_x(x) = nr_x(x) = n(x)$, and $n(x) \leq r_xn(x) \leq nr_xn(x) = n(x)$, so $n(x) \in b(x)$. If $n(x) \in b(x)$ for all x then $n(s) \in b(s) \subseteq S$ for $s \in S$. Open nuclei: $a \to x \in b(x)$. Double negation: $\neg\neg x = \neg x \to x$. Suppose the property holds and a is the least n-dense element. From $nu_b = n \vee u_b = u_bn$ we get $n(b \to x) = b \to n(x)$; with $b = n(x)$ this gives $n(n(x) \to x) = 1$, so $a \leq n(x) \to x$ and $n(x) = (n(x) \to x) \to x \leq a \to x$; and since $n(a) = 1$, $a \to x \leq n(a \to x) = n(a \to x) \wedge n(a) \leq n(x)$. So $n = u_a$. (b) $n \vee m = mn$ iff mn is idempotent; if $\mathrm{Fix}(n)$ is preserved by every m then $mnmn = mmn = mn$, and conversely $mnmn = mn$ at a fixed point t of n gives $nm(t) \leq mnm(t) = m(t)$, so $m(t) \in \mathrm{Fix}(n)$. Taking $m = c_y$ shows that $\mathrm{Fix}(n)$ is an up-set, i.e. n is closed, and closed sublocales are preserved by every nucleus. □

The characterisation transfers to the sheaf topos, because on the subobjects of a sheaf the closure operators compose as the nuclei do.

Corollary 5.12. *Let n, m be nuclei on a frame L and c_n, c_m the corresponding closure operators on* $\mathrm{Sh}(L)$*. Then $c_mc_n \leq c_nc_m$ on the subobjects of every sheaf iff $mn \leq nm$ on L; hence $c_nc_m = c_{n\vee m}$, and $W_n \vee W_m = W_{n\vee m}$ with $k = 1$, iff $nm = n \vee m$. Consequently the topologies j on* $\mathrm{Sh}(L)$ *with $jj' = j \vee j'$ for every topology j' are exactly those whose nucleus satisfies $n(x) \in b(x)$ for all x (the open topologies, the double-negation topology and their joins, which are their composites; only the open ones when the n-dense elements have a least one), and those with $j'j = j \vee j'$ for every j' are exactly the closed topologies.*

Proof. For a subsheaf A of a sheaf X and $s \in X(u)$ put $[\![s \in A]\!] := \bigvee\{v \leq u : s|_v \in A(v)\}$; since A is a subsheaf, the set $\{v \leq u : s|_v \in A(v)\}$ is a down-set closed under joins, so it is the principal down-set of $[\![s \in A]\!]$, and $[\![s|_v \in A]\!] = [\![s \in A]\!] \wedge v$. The topology of n acts on $\Omega(u) = {\downarrow}u$ by $v \mapsto n(v) \wedge u$, so $s \in c_n(A)(u)$ iff $n([\![s \in A]\!]) \geq u$, and

$$[\![s \in c_n(A)]\!] = \bigvee\{v \leq u : n([\![s \in A]\!] \wedge v) \geq v\} = n([\![s \in A]\!]) \wedge u,$$

because $n(a \wedge v) = n(a) \wedge n(v) \geq v$ iff $n(a) \geq v$. Hence $s \in c_mc_n(A)(u)$ iff $m(n([\![s \in A]\!]) \wedge u) \geq u$ iff $mn([\![s \in A]\!]) \geq u$, and likewise for c_nc_m. If $mn \leq nm$, the inclusion $c_mc_n \leq c_nc_m$ follows. Conversely, given $a \in L$, apply it to $A = y(a) \subseteq X = y(mn(a))$ and $s = \mathrm{id}$, for which $[\![s \in A]\!] = a$: $s \in c_mc_n(A)$ as $mn(a) \geq mn(a)$, so $nm(a) \geq mn(a)$. The second assertion is

Lemma 4.4, and the rest is Proposition 5.11, every topology on $\mathrm{Sh}(L)$ coming from a nucleus; two nuclei with the property $n(S) \subseteq S$ for all sublocales have a join $n \vee n' = nn'$ with the same property. □

Here Theorem 4.3 does not apply: in a poset the pushout of $y \hookleftarrow s \hookrightarrow y$ is y with both injections the identity, its fold map is id_y, and no witnesses exist (the only regular epimorphisms are identities, whose kernel pair is the diagonal). Accordingly the criterion is pointwise, as in [21, Thm. 1.1], and the examples there of nuclei on the Gödel chain $[0,1]$ that stabilise pointwise but not uniformly separate $\mathrm{Same}(L)$ from $\mathrm{Same}(\mathrm{Sh}(L))$: with $F = \{0,1\} \cup \{1/2k\}$ and $G = \{0,1\} \cup \{1/(2k+1)\}$ as fixed sets, $W_{n_F} \vee W_{n_G} = W_{n_F \vee n_G}$ in $\mathrm{Same}(L)$ by Corollary 5.10 but not in $\mathrm{Same}(\mathrm{Sh}([0,1]))$ by Corollary 5.9 [23, Ex. 7.4].

5.4. **Fuzzy sets.** Let L be a complete lattice and $\mathbf{Set}(L)$ Goguen's category of L-fuzzy sets: objects (X,μ) with $\mu\colon X \to L$, morphisms $f\colon (X,\mu) \to (Y,\nu)$ maps with $\mu \leq \nu f$. It has finite limits (computed on underlying sets with the initial structure) but is not exact: for the three-element chain $L = \{0 < a < 1\}$ and $X = \{x_0, x_1\}$ with $\mu \equiv 1$, the relation on $X \times X$ with degree 1 on the diagonal and a off it is an internal equivalence relation that is the kernel pair of no map, as kernel pairs of maps out of (X,μ) have degree 1 wherever they are nonempty. For a nucleus j on L preserving binary meets, the j-closed fuzzy sets ($\mu = j\mu$) form a reflective subcategory with reflector $(X,\mu) \mapsto (X, j\mu)$, and the class W_j of maps inverted by it consists of the bijections f with $j\mu = j\nu f$ [21]. Subobjects of (Y,ν) are the (A,α) with $A \subseteq Y$ and $\alpha \leq \nu|_A$, pulled back along f to $(f^{-1}A, \alpha f \wedge \mu)$; put

$$c_j(A,\alpha) := (Y,\ j\bar{\alpha} \wedge \nu),$$

where $\bar{\alpha}$ extends α by the bottom element outside A. This is inflationary, monotone, idempotent, preserves binary intersections (as j preserves binary meets) and is pullback-stable: both $f^{-1}(c_j(A,\alpha))$ and $c_j(f^{-1}(A,\alpha))$ are $(X, j(\bar{\alpha}f) \wedge \mu)$, using $\mu \leq \nu f$ and $\mu \leq j\mu$. A bijection $f\colon (X,\mu) \to (Y,\nu)$ is a monomorphism with image $(Y, \mu f^{-1})$, so it is c_j-dense iff $j(\mu f^{-1}) \geq \nu$ iff $j\mu = j\nu f$: on bijections, c_j-density is membership in W_j, and $f \in M_k$ iff $(jj')^k\mu \geq \nu f$. Bijections form a sameness B, and for a bijection f the inclusions $\mathrm{im}f \hookrightarrow c_j(\mathrm{im}f)$ are bijections, since $c_j(\mathrm{im}f)$ has underlying set Y. Hence Proposition 3.5 gives

$$W_j \vee W_{j'} = \{f \text{ bijective}:\ (jj')^k\mu \geq \nu f \text{ for some } k\},$$

the description of [21, Lemma 4.1, Thm. 4.3] by uniformly bounded alternating chains, each such f being a composite of bijections alternately in $W_{j'}$ and W_j. The uniform criterion $(j'j)^k = j \vee j'$ of [21, Thm. 1.1(ii)] is proved there with witnesses that glue fuzzy sets along a common part and sums of such witnesses, in the pattern of Theorem 4.3; we do not claim $\mathbf{Set}(L)$ as an instance of Theorem 1.2. (For non-bijective maps the class of c_j-bidense maps is larger than W_j: the fold map of two points of degree $\perp$ onto one

is c_j-bidense for every j. The join of two classes of bijections is computed among bijections, so this does not affect the statement.)

5.5. **Abelian categories and torsion theories.** Let $\mathcal{A}$ be a Grothendieck abelian category. A *hereditary torsion theory* is given by a class T of objects closed under subobjects, quotients, extensions and coproducts; equivalently by its radical $t\colon \mathcal{A} \to \mathcal{A}$, the subfunctor of the identity with $t(M)$ the largest subobject of M in T, which is left exact ($t(N) = N \cap t(M)$ for $N \subseteq M$) and idempotent ($t(M/t(M)) = 0$) [5, 22].

Proposition 5.13. *Closure operators on $\mathcal{A}$ correspond bijectively to hereditary torsion theories (this is classical, cf.* [3, 22]*; we include the proof to fix the dictionary): T gives $c_T(A) := \pi^{-1}(t(X/A))$ for $A \subseteq X$ with $\pi\colon X \to X/A$, and c gives $T_c = \{M : c(0 \subseteq M) = M\}$ with radical $t_c(M) = c(0 \subseteq M)$. Under this correspondence $c_T \leq c_{T'}$ iff $T \subseteq T'$, W_{c_T} is the class of maps with kernel and cokernel in T, the maps inverted by the quotient functor $\mathcal{A} \to \mathcal{A}/T$, and the least closure operator above c_T and $c_{T'}$ is $c_{T \vee T'}$ for the least hereditary torsion theory $T \vee T'$ containing T and T'.*

Proof. c_T is inflationary, monotone (the image of $t(X/A)$ in X/B lies in $t(X/B)$ for $A \subseteq B$) and idempotent ($X/c_T(A) = (X/A)/t(X/A)$ is torsion-free). It preserves intersections: $X/(A \cap B)$ embeds in $X/A \oplus X/B$, and as t is left exact and additive, $x + (A \cap B)$ is torsion iff $x + A$ and $x + B$ are. It is pullback-stable: for $f\colon X' \to X$, $X'/f^{-1}(A)$ embeds in X/A, so $t(X'/f^{-1}(A))$ is the preimage of $t(X/A)$ by left exactness, i.e. $c_T(f^{-1}A) = f^{-1}(c_T A)$. Conversely, for a closure operator c, $t_c(M) = c(0)$ is a subfunctor (by Lemma 2.2(i)), left exact (pullback stability along monomorphisms), and idempotent ($c(0 \subseteq M/c(0))$ pulls back to $c(c(0)) = c(0)$), so T_c is a hereditary torsion theory, and $c(A) = \pi^{-1}(c(0 \subseteq X/A)) = c_{T_c}(A)$ by pullback stability along π; and $t_{c_T} = t$. Order: $c_T \leq c_{T'}$ iff $t \leq t'$ iff $T \subseteq T'$. A map $f\colon X \to Y$ has $c_T(\mathrm{im} f) = Y$ iff $Y/\mathrm{im} f = \mathrm{Coker}\, f$ is torsion, and $\ker f \subseteq c_T(\Delta_X)$ iff $\mathrm{Ker}\, f \subseteq t(X)$ iff $\mathrm{Ker}\, f \in T$ by (1); these are the maps inverted by the quotient functor [5, III]. The last assertion follows from the order isomorphism. □

For a word w in $c_T, c_{T'}$ the w-torsion objects are those with a filtration by the canonical subobjects $0 \subseteq t'(M) \subseteq \pi^{-1}t(M/t'(M)) \subseteq \cdots$; so an object is $c^{(k)}$-torsion iff it lies in the class E_k of Corollary 1.3, any alternating filtration being dominated by the canonical one, and E_k is closed under subobjects and quotients. By (1) and pullback stability along $Y \to Y/\mathrm{im} f$, a map is in U_k iff its kernel and cokernel lie in E_k.

Proof of Corollary 1.3. Theorems 1.1 and 1.2 with Proposition 5.13 and the description of U_k; the identity $c^{(k)} = \bar{c}$ says that every $c_{T \vee T'}$-torsion object, i.e. every object of $T \vee T'$, is $c^{(k)}$-torsion. □

By Lemma 4.1, $T \vee T'$ is the class of objects with a transfinite alternating filtration, the classical description of the join of two hereditary torsion

theories; the corollary says that the join of the two classes of maps inverted by the Gabriel quotients is the class of the join torsion theory exactly when a finite bound on the length exists. When T and T' commute in the sense that $t(t'(M)) = t'(t(M))$ and the join is the class of extensions of a T-object by a T'-object, $k = 1$: for instance the p-torsion and the q-torsion abelian groups for distinct primes.

Example 5.14. Let $\mathcal{A}$ be the category of persistence modules over a field k: representations $\cdots \to M_2 \to M_1 \to M_0$ of the ordinal ω with arrows $n+1 \to n$, a Grothendieck abelian category. Let T and T' be the classes of modules supported on the even, respectively on the odd, vertices; each is closed under subobjects, quotients, extensions and direct sums, hence a hereditary torsion theory, with $t(M)_n = 0$ for n odd and $t(M)_n = \mathrm{Ker}(M_n \to M_{n-1})$ for n even ($t(M)_0 = M_0$), and symmetrically t'. A map f has kernel and cokernel in T iff f_n is an isomorphism for every odd n, so W_T is the class of maps that are isomorphisms at the odd vertices and $W_{T'}$ at the even vertices, the linear analogue of the classes of [16, Thm. 7.1]. The join torsion theory is everything: a module that is torsion-free for both has $M_0 = 0$ and all maps $M_n \to M_{n-1}$ injective, hence is 0. The interval module $I = k[0, N]$, with $I_n = k$ for $n \leq N$, identities as maps and $I_n = 0$ for $n > N$, has $t'(I) = 0$, $t(I) = k[0, 0]$, then $t'(I/k[0, 0]) = k[1, 1]$, and so on: its canonical alternating filtration is $0 \subseteq 0 \subseteq k[0, 0] \subseteq k[0, 1] \subseteq \cdots \subseteq k[0, N]$, of length $N + 2$, and $I \in E_k$ iff $2k \geq N + 2$. So $E_k \neq T \vee T'$ for every k, and by Corollary 1.3 the join of "isomorphism at the odd vertices" and "isomorphism at the even vertices" is a proper class of maps, although the join localisation is the zero category and $W_{T\vee T'}$ is the class of all maps; it is the class of maps whose kernel and cokernel have alternating filtrations of bounded length, for instance it contains $0 \to k[0, N]$ for each N but not $0 \to \bigoplus_N k[0, N]$. The witness of Proposition 4.6 for $S = 0 \subseteq k[0, N]$ is the fold map $k[0, N] \oplus k[0, N] \to k[0, N]$, in $W_{T\vee T'}$ but not in U_k for $2k < N+2$, and the direct sum of these fold maps over N lies in no U_k. Neither torsion theory is stable: the injective hull of the simple module $k[n, n]$ is $k[n, \infty)$, supported on both parities.

Stable torsion theories. Recall that a hereditary torsion theory T is *stable* if T is closed under injective hulls [22, Ch. VI, §7]. Parts (i) and (ii) of the following lemma are classical for module categories [22, Ch. VI], and the decomposition of injectives in (iii) is the standard tool for stable theories, cf. [6]; we include the short proofs since we work in a Grothendieck abelian category.

Lemma 5.15. *Let T, T' be hereditary torsion theories on a Grothendieck abelian category $\mathcal{A}$. (i) The torsion-free class F of T is closed under injective hulls. (ii) If T is stable then $t(E)$ is injective, hence a direct summand, for every injective E. (iii) If T and T' are stable then so are $T \cap T'$ and $T \vee T'$.*

Proof. (i) If $N \in F$ and $t(E(N)) \neq 0$ then $t(E(N)) \cap N \neq 0$, and it lies in T as a subobject of a torsion object. (ii) $E(t(E))$ embeds in E and lies in T, so it is contained in $t(E)$. (iii) For E injective, $t_{T\cap T'}(E) = t(t'(E))$ is injective by (ii) applied twice. Write $E = t(E) \oplus E_1$ with E_1 injective and T-torsion-free, and $E_1 = t'(E_1) \oplus E_2$ with E_2 T'-torsion-free; E_2 is also T-torsion-free as a subobject of E_1, hence torsion-free for $T \vee T'$, whose torsion-free class is $F \cap F'$. Then $D := t(E) \oplus t'(E_1)$ lies in $T \vee T'$ and $\bar{t}(E)/D$ embeds in $E/D \cong E_2$, so $\bar{t}(E) = D$ is injective. For $M \in T \vee T'$ the injective subobject $\bar{t}(E(M))$ of $E(M)$ contains the essential subobject M, so $E(M) = \bar{t}(E(M)) \in T \vee T'$. □

Proof of Theorem 1.4. Let $M \in T \vee T'$, $N := M/t(M)$, which is T-torsion-free, and $E := E(N)$. By Lemma 5.15(i) E is T-torsion-free, and $E \in T \vee T'$ because $T \vee T'$ is stable. By Lemma 5.15(ii) for T', $E = t'(E) \oplus E''$ with E'' T'-torsion-free, and E'' is T-torsion-free as a subobject of E; so E'' is torsion-free for $T \vee T'$, while it is a quotient of $E \in T \vee T'$. Hence $E'' = 0$ and $N \subseteq E = t'(E) \in T'$.

For the closure operators let $A \subseteq X$. Since $T \subseteq T \vee T'$, $t(X/A) \subseteq \bar{t}(X/A) = \bar{c}(A)/A$, so $c_T(A) \subseteq \bar{c}(A)$ and $c_T(A)$ is the c_T-closure of A computed in $\bar{c}(A)$; likewise $c_{T'}(c_T(A)) \subseteq \bar{c}(A)$ is the $c_{T'}$-closure of $c_T(A)$ computed in $\bar{c}(A)$. As $\bar{c}(A)/A \in T \vee T'$, the first part gives $\bar{c}(A)/c_T(A) = (\bar{c}(A)/A)/t(\bar{c}(A)/A) \in T'$, i.e. $c_{T'}(c_T(A)) = \bar{c}(A)$. Thus $c'c = \bar{c}$ in the notation of Section 4, so $(cc')^2 = \bar{c}$ and the criterion of Theorem 1.2 holds; every object of $T \vee T'$ has the filtration $0 \subseteq t(M) \subseteq M$, which is the filtration of Corollary 1.3 with the roles of T and T' exchanged and $k = 1$. The four-map factorisation is the peeling of Lemma 3.3: for $f \in W_{T\vee T'}$ with kernel K and cokernel C, the maps $X \to X/t(K)$ and $X/t(K) \to X/K \cong \mathrm{im} f$ have kernels $t(K) \in T$ and $K/t(K) \in T'$, and with $Y_1 \subseteq Y$ the preimage of $t(C)$ the inclusions $\mathrm{im} f \subseteq Y_1 \subseteq Y$ have cokernels $t(C) \in T$ and $C/t(C) \in T'$; the composite of the four is f, and the closure of $W_T \cup W_{T'}$ under composition is the join by Theorem 1.1. Proposition 5.17 below reduces the four maps to two. If both theories are stable, $T \vee T'$ is stable by Lemma 5.15(iii), the argument applies to both orders, and $c_T c_{T'} = \bar{c} = c_{T'} c_T$. □

For the two-step factorisation we use the localisation functor. Let $q_T \colon \mathcal{A} \to \mathcal{A}/T$ be Gabriel's exact quotient functor and $s_T \colon \mathcal{A}/T \to \mathcal{A}$ its fully faithful right adjoint, which exists because $\mathcal{A}$ is a Grothendieck category and T is closed under coproducts [5, Ch. III], [22, Ch. IX–X]; put $Q_T := s_T q_T$ with unit $\ell_M \colon M \to Q_T(M)$. We collect the facts we need, in the form valid in any Grothendieck abelian category.

Lemma 5.16. *(i) Q_T is left exact, and $Q_T(N) = 0$ for $N \in T$; hence $Q_T(M) \cong Q_T(M/t(M))$ for every M. (ii)* $\mathrm{Ker}\, \ell_M = t(M)$ *and* $\mathrm{Coker}\, \ell_M \in T$. *(iii) If M is T-torsion-free then ℓ_M is a monomorphism and $Q_T(M)$ embeds in the injective hull $E(M)$. (iv) Q_T has right derived functors*

R^nQ_T, computable from any injective resolution, with the usual long exact sequences.

Proof. (i) q_T is exact and s_T, a right adjoint, is left exact; q_T kills T, and applying Q_T to $0 \to t(M) \to M \to M/t(M) \to 0$ gives $Q_T(M) \cong Q_T(M/t(M))$ by exactness of q_T and full faithfulness of s_T. (ii) is in [5, Ch. III] and [22, Ch. IX, §1]: $\operatorname{Ker}\ell_M$ and $\operatorname{Coker}\ell_M$ are killed by q_T, i.e. lie in T, and $\operatorname{Ker}\ell_M \supseteq t(M)$ while $\operatorname{Ker}\ell_M/t(M)$ embeds in $Q_T(M)$, which is torsion-free. (iii) By Lemma 5.15(i) $E(M)$ is torsion-free, and being injective it is T-closed (torsion-free with $\operatorname{Ext}^1(N, E(M)) = 0$ for $N \in T$), so $\ell_{E(M)}$ is an isomorphism [22, Ch. IX, §1]; applying the left exact Q_T to $M \hookrightarrow E(M)$ gives $Q_T(M) \hookrightarrow Q_T(E(M)) \cong E(M)$, and ℓ_M is a monomorphism by (ii). (iv) $\mathcal{A}$ has enough injectives and Q_T is left exact. □

Proposition 5.17. *Let T, T' be hereditary torsion theories on a Grothendieck abelian category with T' and $T \vee T'$ stable.*

(i) *If $N \in T'$ then $Q_T(N) \in T'$ and $R^nQ_T(N) \in T'$ for all $n \geq 0$; if $C \in T \vee T'$ then $Q_T(C) \in T'$.*
(ii) *Let $f\colon X \to Y$ lie in $W_{T\vee T'}$ and have T-torsion-free kernel, let $Z := Y \times_{Q_T(Y)} Q_T(X)$ be the pullback of ℓ_Y along $Q_T(f)$, with projections $h\colon Z \to Y$ and $p\colon Z \to Q_T(X)$, and let $g\colon X \to Z$ be the map with $hg = f$ and $pg = \ell_X$. Then $g \in W_T$ and $h \in W_{T'}$.*
(iii) *$W_{T\vee T'} = W_{T'} \circ W_T$: every map inverted by $Q_{T\vee T'}$ is a map inverted by Q_T followed by a map inverted by $Q_{T'}$.*

Proof. (i) By Lemma 5.16(i),(iii), $Q_T(N) \cong Q_T(N/t(N))$ is a subobject of $E(N/t(N))$, which lies in T' since $N/t(N) \in T'$ and T' is stable; so $Q_T(N) \in T'$. As T' is stable, N has an injective resolution $0 \to N \to E^0 \to E^1 \to \cdots$ with all $E^n \in T'$ (take $E^0 = E(N)$, then the hull of $E^0/N \in T'$, and so on), and $R^nQ_T(N)$ is a subquotient of $Q_T(E^n) \in T'$ by Lemma 5.16(iv). If $C \in T \vee T'$ then $C/t(C) \in T'$ by Theorem 1.4 and $Q_T(C) = Q_T(C/t(C)) \in T'$.

(ii) Let K, I and C be the kernel, image and cokernel of f; $K \in T'$ since $K \in T \vee T'$ is T-torsion-free, and $C \in T \vee T'$. The map h is the pullback of $Q_T(f)$ along ℓ_Y, so $\operatorname{Ker}h \cong \operatorname{Ker}Q_T(f) = Q_T(K)$ (Lemma 5.16(i)), which lies in T' by (i), and $\operatorname{Coker}h$ embeds in $C' := \operatorname{Coker}Q_T(f)$. The sequence $0 \to K \to X \to I \to 0$ gives the exact sequence $Q_T(X) \to Q_T(I) \to R^1Q_T(K)$, so $Q_T(I)/\mathrm{im}Q_T(f)$ embeds in $R^1Q_T(K) \in T'$; the sequence $0 \to I \to Y \to C \to 0$ gives $Q_T(Y)/Q_T(I) \hookrightarrow Q_T(C) \in T'$. Hence C' is an extension of two objects of T', $C' \in T'$, and $h \in W_{T'}$. The map p is the pullback of ℓ_Y along $Q_T(f)$, so $\operatorname{Ker}p \cong t(Y) \in T$ and $\operatorname{Coker}p$ embeds in $\operatorname{Coker}\ell_Y \in T$. Since $pg = \ell_X$ has kernel $t(X)$ and cokernel in T (Lemma 5.16(ii)), the exact sequence

$$0 \to \operatorname{Ker}g \to \operatorname{Ker}\ell_X \to \operatorname{Ker}p \to \operatorname{Coker}g \to \operatorname{Coker}\ell_X \to \operatorname{Coker}p \to 0$$

of the composite shows that $\operatorname{Ker}g$ and $\operatorname{Coker}g$ lie in T, i.e. $g \in W_T$.

(iii) The inclusion $W_{T'} \circ W_T \subseteq W_{T \vee T'}$ is clear. For $f\colon X \to Y$ in $W_{T \vee T'}$ with kernel K, the quotient map $q\colon X \to X/t(K)$ lies in W_T and $f = f'q$ with $f' \in W_{T \vee T'}$ having the T-torsion-free kernel $K/t(K)$; by (ii) $f' = hg$ with $g \in W_T$, $h \in W_{T'}$, and $f = h(gq)$ with $gq \in W_T$. □

Remark 5.18. For arbitrary hereditary torsion theories the two-step factorisation implies the first assertion of Theorem 1.4: if $W_{T \vee T'} = W_{T'} \circ W_T$ and $M \in T \vee T'$, write $0 \to M$ as $h \circ g$ with $g\colon 0 \to Z$ in W_T, so $Z \in T$, and $h\colon Z \to M$ in $W_{T'}$; then $h(Z) \in T$ and $M/h(Z) = \operatorname{Coker} h \in T'$, so $M/t(M) \in T'$. Thus $W_{T \vee T'} = W_{T'} \circ W_T$ iff $c_{T'}c_T = \bar{c}$ whenever T' and $T \vee T'$ are stable, and the implication from left to right holds in general; the converse fails without stability (Example 5.19), and Example 5.8 is the analogous failure for topologies.

Example 5.19. Let Λ be the path algebra over a field k of the quiver $1 \xrightarrow{a} 2 \xrightarrow{b} 3$ with the relation $ba = 0$, let S_i be the simple representation at the vertex i, and let $T = \operatorname{Add} S_1$ and $T' = \operatorname{Add} S_3$, the classes of all direct sums of copies of S_1, resp. of S_3, be the hereditary torsion theories of the representations supported at 1, resp. at 3. A representation with $M_2 = 0$ is $S_1^{(I)} \oplus S_3^{(J)}$, so $T \vee T'$ is the class of these, $M/t(M) \in T'$ and $M/t'(M) \in T$ for all $M \in T \vee T'$, and $c_{T'}c_T = \bar{c} = c_T c_{T'}$: the closure operators commute, and the criterion holds with $k = 1$ in both orders. Nevertheless $W_{T \vee T'} \neq W_{T'} \circ W_T$. Because of the relation, the projective $P_2 = (0 \to k \to k)$ is the injective hull of S_3 and $P_1 = (k \to k \to 0)$ is the injective hull of S_2; let $m\colon P_2 \to P_1$ be the composite $P_2 \twoheadrightarrow S_2 \hookrightarrow P_1$, with kernel $S_3 \in T'$ and cokernel $S_1 \in T$, so that $m \in W_{T \vee T'}$. Suppose $m = hg$ with $g\colon P_2 \to Z$ in W_T and $h\colon Z \to P_1$ in $W_{T'}$. As $\operatorname{Hom}(S_1, P_2) = 0$, g is a monomorphism with cokernel in $\operatorname{Add} S_1$, and as P_2 is injective, $Z \cong P_2 \oplus S_1^{(I)}$; as $\operatorname{Hom}(S_1, P_1) = 0$, h vanishes on $S_1^{(I)}$, so $S_1^{(I)} \subseteq \operatorname{Ker} h \in T'$ forces $I = \emptyset$, $Z = P_2$ and $h = m$, whose cokernel S_1 is not in T'. Hence $m \notin W_{T'} \circ W_T$; here T' is not stable, $E(S_3) = P_2 \notin T'$. In the other order $m = (S_2 \hookrightarrow P_1) \circ (P_2 \twoheadrightarrow S_2)$ lies in $W_T \circ W_{T'}$. Example 5.8 is the analogue for topologies. Here the localisation Q_T does not preserve epimorphisms, the cokernel of $Q_T(X) \to Q_T(\operatorname{im} f)$ embedding in $R^1Q_T(\operatorname{Ker} f)$; there the c'-sheafification preserves the epimorphisms of sheaves but not the c-bidense epimorphisms of the topos.

Corollary 5.20. *Let R be a commutative Noetherian ring. For a specialisation-closed subset $Y \subseteq \operatorname{Spec} R$ let T_Y be the class of modules with support in Y and $W_Y := W_{T_Y}$, the class of homomorphisms f whose localisation $f_\mathfrak{p}$ is an isomorphism for every prime $\mathfrak{p} \notin Y$, i.e. whose associated map of quasi-coherent sheaves is an isomorphism over* $\operatorname{Spec} R \setminus Y$. *Then*

$$W_Y \vee W_{Y'} = W_{Y \cup Y'} = W_{Y'} \circ W_Y, \qquad W_Y \cap W_{Y'} = W_{Y \cap Y'},$$

so $Y \mapsto W_Y$ is a lattice embedding of the lattice of specialisation-closed subsets of $\operatorname{Spec} R$ *into* $\operatorname{Same}(\operatorname{Mod} R)$.

Proof. By Gabriel's theorem the hereditary torsion theories on $\operatorname{Mod} R$ are exactly the classes T_Y, and the join of T_Y and $T_{Y'}$ is $T_{Y\cup Y'}$ [22, Ch. VII]; W_Y is as described because $\operatorname{Supp}(\operatorname{Ker} f) \cup \operatorname{Supp}(\operatorname{Coker} f) \subseteq Y$ iff $f_{\mathfrak{p}}$ is an isomorphism for all $\mathfrak{p} \notin Y$. Every T_Y is stable: every element of the injective hull $E(R/\mathfrak{p})$ is annihilated by a power of $\mathfrak{p}$, so $E(R/\mathfrak{p}) \in T_Y$ if $\mathfrak{p} \in Y$; if $\mathfrak{p} \notin Y$, every nonzero submodule of $E(R/\mathfrak{p})$ has $\mathfrak{p}$ as an associated prime, hence a support not contained in Y, so $E(R/\mathfrak{p})$ is T_Y-torsion-free; and since every injective module is a direct sum of such hulls and t_Y commutes with direct sums, $t_Y(E)$ is a direct sum of injectives, hence injective as R is Noetherian, so $E(M) = t_Y(E(M))$ for $M \in T_Y$. Theorem 1.4 and Proposition 5.17 give the join; for the meet, a map has kernel and cokernel in $T_Y \cap T_{Y'} = T_{Y\cap Y'}$ iff it lies in $W_Y \cap W_{Y'}$. □

In this case the first assertion of Theorem 1.4 has a direct proof: if $\operatorname{Supp} M \subseteq Y \cup Y'$ then no associated prime of $M/t_Y(M)$ lies in Y, and since over a Noetherian ring every prime in the support of a module contains an associated prime, $\operatorname{Supp}(M/t_Y(M)) \subseteq Y'$. For $Y = V(I)$ and $Y' = V(J)$ the corollary says that a homomorphism inducing an isomorphism of sheaves over $D(I)\cap D(J) = D(IJ)$ is the composite of a homomorphism inducing an isomorphism over $D(I)$ followed by one inducing an isomorphism over $D(J)$; the intermediate module is the pullback of the target along the localisation of the source at $D(I)$.

Example 5.21. Let Q be the quiver with vertices $1, 2$ and arrows $a\colon 1 \to 2$, $b\colon 2 \to 1$, and $\Lambda = kQ$ its path algebra over a field k, a Noetherian ring (it is finitely generated as a module over its central subalgebra $k[ab + ba]$). A Λ-module is a pair of vector spaces $M_1 \rightleftarrows M_2$; let T be the class of modules with $M_2 = 0$ and T' that with $M_1 = 0$, hereditary torsion theories with $t(M) = (\operatorname{Ker} a, 0)$ and $t'(M) = (0, \operatorname{Ker} b)$. For $N \geq 1$ let M_N be the string module with basis $e_1, \dots, e_N$, the e_i placed alternately at the vertices $1, 2, 1, \dots$, each arrow sending e_i to e_{i+1} and e_N to 0. It is uniserial with composition factors the simple modules S_1, S_2 at the two vertices, so $M_N \in T \vee T'$, and its canonical alternating filtration, which begins with $t'(M_N)$ and peels off one basis vector at a time, has length N if N is even and $N + 1$ if N is odd, when e_N lies at the vertex 1 and the first step $t'(M_N) = 0$ is trivial (lengths are counted as in Example 5.14, trivial steps included). Hence $T \vee T' \not\subseteq E_k$ for every k, and $W_T \vee W_{T'} \subsetneq W_{T\vee T'}$ although Λ is Noetherian; the map $0 \to \bigoplus_N M_N$ lies in the second class but not in the first. In accordance with Theorem 1.4, T is not stable: M_N with e_N at the vertex 1 is an essential extension of its socle $S_1 = ke_N$ and does not lie in T, so neither does $E(S_1)$. (The join class is not everything: the module $k \rightleftarrows k$ with both arrows the identity is simple and torsion-free for both theories.) A computer check over $\mathbb{F}_2$ finds, among the modules of dimension vector (d, d) with $d \leq 5$ in the join class, the maximal length $2d + 1$, attained by the string module with $2d$ basis vectors whose first vector lies at the vertex 2, so that its socle ke_{2d} is S_1 and its first step is trivial; M_{2d} itself has length $2d$.

For the quiver A_2 (the case $b = 0$) one has $t'(M) = (0, M_2)$ with quotient in T, so every module lies in E_1 although T' is not stable ($E(S_2) = (k \to k)$): stability is sufficient, not necessary. The order of the layers matters there, since $c_{T'}c_T \neq \bar{c}$: for $M = (k \to k)$ with a an isomorphism, $t(M) = 0$ and $M/t'(M) = S_1 \notin T'$; this is what Theorem 1.4 applied to the pair (T', T) predicts, T being stable ($E(S_1) = S_1$). The order in Proposition 5.17(iii) is forced as well: for A_2 one has $W_{T\vee T'} = W_T \circ W_{T'}$ (the localisation at T' is $M \mapsto (M_1 \to 0)$, and the pullback factorisation of a map f has middle term $(X_1 \to Y_2)$), but $0 \to (k \xrightarrow{\cong} k)$ is not a map of W_T followed by a map of $W_{T'}$: the middle term would lie in T, and a map $(Z_1 \to 0) \to (k \to k)$ is zero at the vertex 1, hence not an isomorphism there.

5.6. **Summary.** The settings are instances of one theorem, and the identity $(cc')^k = \bar{c}$ that governs them is, for nuclei, the finite-iteration axiom of the lax logic of the join of two nuclei [19]: uniform in the subobject in toposes, locales, abelian categories and fuzzy sets, where witnesses and sums exist, and pointwise in the frame itself, where they do not. The pointwise condition is also the one that characterises $\Box_C \vee \Box_{C'} = \Box_{C\cap C'}$ for relativized necessities on the frame of up-sets of a poset (absence of infinite alternating chains [20]), and a disjoint union of finite alternating chains of unbounded length satisfies it but not the uniform one.

6. The dual regime: quotient-closed reflections

Let $\mathcal{V}$ be a variety of algebras and $R \subseteq \mathcal{V}$ a subvariety, with reflector $a_R(A) = A/\rho_R(A)$, $\rho_R(A)$ the least congruence with quotient in R. A homomorphism $f\colon A \to B$ is inverted by a_R iff $\mathrm{im}f$ meets every $\rho_R(B)$-class and $(f \times f)^{-1}(\rho_R(B)) \subseteq \rho_R(A)$. This looks like bidensity, but the candidate closure operators are not pullback-stable: for R the abelian groups, $\rho_R(A)$ is the commutator congruence, and the inclusion $A_3 \hookrightarrow S_3$ has $\rho_R(A_3)$ trivial while $\rho_R(S_3)$ restricted to A_3 is everything; likewise the $\rho_R(B)$-saturation of a subalgebra is not stable under intersection with subalgebras (in S_3, with $K = \{e, (13)\}$ and $A = \{e, (12)\}$, the saturation of A contains (13) but $A \cap K$ is trivial). Accordingly the mechanism is different: because R is closed under quotients, the composite $a_{R'}a_R$ is already the reflector onto $R \cap R'$, and the join theorem of [15, Thm. 4.1] gives $W_R \vee W_{R'} = W_{R\cap R'}$ for all subvarieties, with no condition [18, Thm. 1.1]: at the level of reflectors, $k = 1$. The hard direction there is the meet, $W_R \cap W_{R'} = W_{R\vee R'}$, which holds iff a fracture condition is satisfied (subdirect representation and absence of dense proper subalgebras [18, Thm. 1.2]). The two regimes are thus dual: for closure-type reflections (subtoposes) the meet of classes is automatic and the join is governed by finite iteration of closures; for quotient-closed reflections (subvarieties) the join is automatic and the meet is governed by fracture. Remark 5.2 shows that in the closure-type regime the reflector-level condition ($k = 1$ or any uniform k for the tower of reflectors) is not the right one.

7. Questions

Question 7.1. Theorem 1.2 reduces join preservation to the identity $(cc')^k = \bar{c}$. For subspaces this is the uniform termination of the alternating derivation on closed sets, for ideal topologies the stabilisation of the ideal chain. Is there a structural description of the pairs of nuclei, or of subtoposes, satisfying it, for instance in terms of the lattice of sublocales?

Question 7.2. Which pairs of non-stable hereditary torsion theories, beyond those of the quiver A_2 in Example 5.21, have bounded alternation, and which of them have the two-step factorisation, which the identity $c_{T'}c_T = \bar{c}$ alone does not give (Example 5.19)? Likewise, for which pairs of topologies with $c'c \leq cc'$ does the c-sheafification preserve c'-sheaves, so that $W_{\bar{c}} = W_c \circ W_{c'}$ (Proposition 5.5)? Not for all commuting pairs (Example 5.8); in a search over all Grothendieck topologies on the categories with two objects and at most six morphisms, testing random c'-sheaves, the preservation failed only for commuting pairs. Which topologies on a non-localic topos satisfy $cc' = c \vee c'$ for every c' (the condition of Proposition 5.4(i) is sufficient, and in a localic topos the property is decided by the nucleus, Corollary 5.12)? The lattice of hereditary torsion theories is itself a frame, and nuclei on it are studied in [8]; the relation between those nuclei and the closure operators considered here is not clear to us.

Question 7.3. Does Theorem 1.2 hold in an elementary topos, where (H1) and the coproducts in (T) may be unavailable? By Remark 4.7 the obstacle is the existence of $\bar{c}$ and the combination of infinitely many witnesses. Likewise, which exact categories beyond toposes and abelian categories have witnesses and sums?

Question 7.4. Is there a common generalisation of the closure-type and the quotient-closed regimes, for reflective subcategories of an exact category whose reflector is determined by a closure operator that is only *weakly* hereditary in the sense of [3]?

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